

NCERT Solutions for Class 12

Maths

Chapter 4 - Determinants

Exercise 4.1

1. Evaluate the determinant: $\begin{vmatrix} 2 & 4 \\ -5 & -1 \end{vmatrix}$.

Ans: Solving the determinant $\begin{vmatrix} 2 & 4 \\ -5 & -1 \end{vmatrix}$, we have:

$$\Rightarrow \begin{vmatrix} 2 & 4 \\ -5 & -1 \end{vmatrix} = 2(-1) - 4(-5)$$

$$\Rightarrow \begin{vmatrix} 2 & 4 \\ -5 & -1 \end{vmatrix} = -2 + 20$$

$$\therefore \begin{vmatrix} 2 & 4 \\ -5 & -1 \end{vmatrix} = 18$$

2. Evaluate the determinants.

i. $\begin{vmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{vmatrix}$

Ans: Solving the determinant $\begin{vmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{vmatrix}$, we have:

$$\Rightarrow \begin{vmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{vmatrix} = (\cos \theta)(\cos \theta) - (-\sin \theta)(\sin \theta)$$

$$\Rightarrow \begin{vmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{vmatrix} = \cos^2 \theta + \sin^2 \theta$$

We know,

$$\cos^2 \theta + \sin^2 \theta = 1$$

$$\therefore \begin{vmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{vmatrix} = 1$$

$$\text{ii. } \begin{vmatrix} x^2 - x + 1 & x - 1 \\ x + 1 & x + 1 \end{vmatrix}$$

Ans: Solving the determinant $\begin{vmatrix} x^2 - x + 1 & x - 1 \\ x + 1 & x + 1 \end{vmatrix}$, we have:

$$\Rightarrow \begin{vmatrix} x^2 - x + 1 & x - 1 \\ x + 1 & x + 1 \end{vmatrix} = (x^2 - x + 1)(x + 1) - (x - 1)(x + 1)$$

$$\Rightarrow \begin{vmatrix} x^2 - x + 1 & x - 1 \\ x + 1 & x + 1 \end{vmatrix} = x^3 - x^2 + x + x^2 - x + 1 - (x^2 - 1)$$

So,

$$\begin{vmatrix} x^2 - x + 1 & x - 1 \\ x + 1 & x + 1 \end{vmatrix} = x^3 + 1 - x^2 + 1$$

$$\therefore \begin{vmatrix} x^2 - x + 1 & x - 1 \\ x + 1 & x + 1 \end{vmatrix} = x^3 - x^2 + 2$$

3. If $A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$, then show that $|2A| = 4|A|$.

Ans: Given that, $A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$

Multiplying A by 2, we have:

$$\Rightarrow 2A = 2 \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 8 & 4 \end{bmatrix}$$

$$\Rightarrow 2A = \begin{bmatrix} 2 & 4 \\ 8 & 4 \end{bmatrix}$$

$$\therefore \text{L.H.S} = |2A| = \begin{vmatrix} 2 & 4 \\ 8 & 4 \end{vmatrix}$$

$$\Rightarrow |2A| = 2 \times 4 - 4 \times 8$$

$$\Rightarrow |2A| = 8 - 32$$

$$\therefore |2A| = -24$$

The value of determinant A is

$$\Rightarrow |A| = \begin{vmatrix} 1 & 2 \\ 4 & 2 \end{vmatrix}$$

$$\Rightarrow |A| = 2 - 8$$

$$\therefore |A| = -6$$

R.H.S is given as $4|A|$.

$$\therefore 4|A| = 4 \times (-6) = -24$$

Hence, we have L.H.S = R.H.S

$$\therefore |2A| = 4|A|.$$

4. If $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix}$, then show that $|3A| = 27|A|$.

Ans: Given, $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix}$

Determining the value of determinant A, by expanding along the first column, i.e., C_1 , we get:

$$\Rightarrow |A| = 1 \begin{vmatrix} 1 & 2 \\ 0 & 4 \end{vmatrix} - 0 \begin{vmatrix} 0 & 1 \\ 0 & 4 \end{vmatrix} + 0 \begin{vmatrix} 0 & 1 \\ 1 & 2 \end{vmatrix}$$

$$\Rightarrow |A| = 1(4 - 0) - 0 + 0$$

$$\therefore |A| = 4$$

Hence, $27|A| = 27 \times 4$

$$\Rightarrow 27|A| = 108 \quad \dots\dots(1)$$

The value of $|3A|$ is obtained as:

$$\Rightarrow 3A = 3 \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix}$$

$$\Rightarrow 3A = \begin{bmatrix} 3 & 0 & 3 \\ 0 & 3 & 6 \\ 0 & 0 & 12 \end{bmatrix}$$

$$\therefore |3A| = 3 \begin{vmatrix} 3 & 6 \\ 0 & 12 \end{vmatrix} - 0 \begin{vmatrix} 0 & 3 \\ 0 & 12 \end{vmatrix} + 0 \begin{vmatrix} 0 & 3 \\ 3 & 6 \end{vmatrix}$$

$$\Rightarrow 3(36-0)+0+0$$

$$\Rightarrow |3A|=3 \times 36$$

$$\text{Thus, } |3A|=108 \dots\dots(2)$$

From equations (1) and (2), we have:

$$|3A|=27|A|$$

Hence proved.

5. Evaluate the determinants

i.
$$\begin{vmatrix} 3 & -1 & -2 \\ 0 & 0 & -1 \\ 3 & -5 & 0 \end{vmatrix}$$

Ans: Let $A = \begin{vmatrix} 3 & -1 & -2 \\ 0 & 0 & -1 \\ 3 & -5 & 0 \end{vmatrix}$

Determining the value of A by expanding along the third row, we have:

$$\Rightarrow |A| = 3 \begin{vmatrix} -1 & -2 \\ 0 & -1 \end{vmatrix} - (-5) \begin{vmatrix} 3 & -2 \\ 0 & -1 \end{vmatrix} + 0 \begin{vmatrix} 3 & -1 \\ 0 & 0 \end{vmatrix}$$

$$\Rightarrow |A| = (3-15)$$

$$\therefore |A| = -12$$

ii.
$$\begin{vmatrix} 3 & -4 & 5 \\ 1 & 1 & -2 \\ 2 & 3 & 1 \end{vmatrix}$$

Ans: Let $A = \begin{vmatrix} 3 & -4 & 5 \\ 1 & 1 & -2 \\ 2 & 3 & 1 \end{vmatrix}$

Determining the value of A by expanding along the first row, we have:

$$\Rightarrow |A| = 3 \begin{vmatrix} 1 & -2 \\ 3 & 1 \end{vmatrix} + 4 \begin{vmatrix} 1 & -2 \\ 2 & 1 \end{vmatrix} + 5 \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix}$$

$$\Rightarrow |A| = 3(1+6) + 4(1+4) + 5(3-2)$$

$$\Rightarrow |A| = 21 + 20 + 5$$

$$\therefore |A| = 46$$

iii. $\begin{vmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ -2 & 3 & 0 \end{vmatrix}$.

Ans: Let $A = \begin{vmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ -2 & 3 & 0 \end{vmatrix}$

Determining the value of A by expanding along the first row, we have:

$$\Rightarrow |A| = 0 \begin{vmatrix} -1 & -3 \\ -2 & 0 \end{vmatrix} - 1 \begin{vmatrix} -1 & -3 \\ -2 & 0 \end{vmatrix} + 2 \begin{vmatrix} -1 & 0 \\ -2 & 3 \end{vmatrix}$$

$$\Rightarrow |A| = 0(9) - (-6) + 2(-3)$$

$$\therefore |A| = 0$$

iv. $\begin{vmatrix} 2 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{vmatrix}$

Ans: Let $A = \begin{vmatrix} 2 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{vmatrix}$

Determining the value of A by expanding along the first column, we have:

$$\Rightarrow |A| = 2 \begin{vmatrix} -1 & -2 \\ -5 & 0 \end{vmatrix} - 0 \begin{vmatrix} -1 & -2 \\ -5 & 0 \end{vmatrix} + 3 \begin{vmatrix} -1 & -2 \\ 2 & -1 \end{vmatrix}$$

$$\Rightarrow |A| = 2(-5) - 0 + 3(5)$$

$$\Rightarrow |A| = -10 + 15$$

$$\therefore |A| = 5$$

6. If $A = \begin{bmatrix} 1 & 1 & -2 \\ 2 & 1 & -3 \\ 5 & 4 & -9 \end{bmatrix}$, find $|A|$.

Ans: Given, $A = \begin{bmatrix} 1 & 1 & -2 \\ 2 & 1 & -3 \\ 5 & 4 & -9 \end{bmatrix}$

Determining the value of A by expanding along the first row, we have:

$$\Rightarrow |A| = 1 \begin{vmatrix} -3 & -9 \\ 4 & -9 \end{vmatrix} - 1 \begin{vmatrix} 2 & -3 \\ 5 & -9 \end{vmatrix} + 2 \begin{vmatrix} 2 & 1 \\ 5 & 4 \end{vmatrix}$$

$$\Rightarrow |A| = 1(-9 + 12) - 1(-18 + 15) - 2(8 - 5)$$

$$\Rightarrow |A| = 3 + 3 - 6$$

$$\therefore |A| = 0$$

7. Find values of x , if

i. $\begin{vmatrix} 2 & 4 \\ 5 & 1 \end{vmatrix} = \begin{vmatrix} 2x & 4 \\ 6 & x \end{vmatrix}$

Ans: Given, $\begin{vmatrix} 2 & 4 \\ 5 & 1 \end{vmatrix} = \begin{vmatrix} 2x & 4 \\ 6 & x \end{vmatrix}$

Solving it, we have:

$$\Rightarrow (2 \times 1) - (5 \times 4) = (2x \times x) - (6 \times 4)$$

$$\Rightarrow 2 - 20 = 2x^2 - 24$$

$$\Rightarrow -18 + 24 = 2x^2$$

$$\Rightarrow 3 = x^2$$

Applying square root on both the sides, we obtain:

$$\Rightarrow x = \pm\sqrt{3}$$

ii. $\begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = \begin{vmatrix} x & 3 \\ 2x & 5 \end{vmatrix}$

Ans: Given, $\begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = \begin{vmatrix} x & 3 \\ 2x & 5 \end{vmatrix}$

Solving it, we have:

$$\Rightarrow (2 \times 5) - (3 \times 4) = (x \times 5) - (3 \times 2x)$$

$$\Rightarrow 10 - 12 = 5x - 6x$$

$$\Rightarrow -2 = -x$$

Multiplying by (-1) on both the sides, we obtain:

$$\Rightarrow x = 2$$

8. If $\begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix}$, then x is equal to

- A. 6
- B. ± 6
- C. -6
- D. 0

Ans: Given, $\begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix}$

Solving it, we have:

$$\Rightarrow x^2 - 36 = 36 - 36$$

$$\Rightarrow x^2 - 36 = 0$$

$$\Rightarrow x^2 = 36$$

Applying square root on both the sides, we obtain:

$$\Rightarrow x = \pm 6$$

Hence, **B. ± 6** is the correct answer.

Exercise 4.2

1. Using the property of determinants and without expanding, prove that:

$$\begin{vmatrix} x & a & x+a \\ y & b & y+b \\ z & c & z+c \end{vmatrix} = 0.$$

Ans: Given matrix $\begin{vmatrix} x & a & x+a \\ y & b & y+b \\ z & c & z+c \end{vmatrix}.$

Applying the Sum Property of determinants, we have

$$\begin{vmatrix} x & a & x+a \\ y & b & y+b \\ z & c & z+c \end{vmatrix} = \begin{vmatrix} x & a & x \\ y & b & y \\ z & c & z \end{vmatrix} + \begin{vmatrix} x & a & a \\ y & b & b \\ z & c & c \end{vmatrix}$$

We know, if two rows or columns of a determinant are identical, then the value of the determinant is zero.

Since, the two columns in both the determinants are identical, thus its determinant would be zero.

$$\Rightarrow \begin{vmatrix} x & a & x+a \\ y & b & y+b \\ z & c & z+c \end{vmatrix} = 0 + 0$$

$$\therefore \begin{vmatrix} x & a & x+a \\ y & b & y+b \\ z & c & z+c \end{vmatrix} = 0$$

2. Using the property of determinants and without expanding, prove that:

$$\begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix} = 0$$

Ans: Let $\Delta = \begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix}$

Applying row operation, $R_1 \rightarrow R_1 + R_2$

$$\Rightarrow \Delta = \begin{vmatrix} a-b+b-c & b-c+c-a & c-a+a-b \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix}$$

$$\Rightarrow \Delta = \begin{vmatrix} a-c & b-a & c-b \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix}$$

$$\Rightarrow \Delta = \begin{vmatrix} a-c & b-a & c-b \\ b-c & c-a & a-b \\ -(a-c) & -(b-a) & -(c-b) \end{vmatrix}$$

Multiplying the third row by (-1) , we get:

$$\Rightarrow \Delta = \begin{vmatrix} a-c & b-a & c-b \\ b-c & c-a & a-b \\ a-c & b-a & c-b \end{vmatrix}$$

We know, if two rows or columns of a determinant are identical, then the value of the determinant is zero.

Since, the two rows R_1 and R_3 are identical.

$$\therefore \Delta = 0$$

Hence, $\begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix} = 0.$

3. Using the property of determinants and without expanding, prove that:

$$\begin{vmatrix} 2 & 7 & 65 \\ 3 & 8 & 75 \\ 5 & 9 & 86 \end{vmatrix} = 0$$

Ans: Let $\Delta = \begin{vmatrix} 2 & 7 & 65 \\ 3 & 8 & 75 \\ 5 & 9 & 86 \end{vmatrix}$

$$\therefore \Delta = \begin{vmatrix} 2 & 7 & 63+2 \\ 3 & 8 & 72+3 \\ 5 & 9 & 81+5 \end{vmatrix}$$

Applying the Sum Property of determinants, we get

$$\Rightarrow \Delta = \begin{vmatrix} 2 & 7 & 63 \\ 3 & 8 & 72 \\ 5 & 9 & 81 \end{vmatrix} + \begin{vmatrix} 2 & 7 & 2 \\ 3 & 8 & 3 \\ 5 & 9 & 5 \end{vmatrix}$$

The two columns of the second determinant are identical, thus its value becomes zero.

Hence,

$$\Rightarrow \Delta = \begin{vmatrix} 2 & 7 & 63 \\ 3 & 8 & 72 \\ 5 & 9 & 81 \end{vmatrix} + 0$$

$$\Rightarrow \Delta = \begin{vmatrix} 2 & 7 & 9(7) \\ 3 & 8 & 9(8) \\ 5 & 9 & 9(9) \end{vmatrix}$$

Taking 9 common from the third column, we have

$$\Rightarrow \Delta = 9 \begin{vmatrix} 2 & 7 & 7 \\ 3 & 8 & 8 \\ 5 & 9 & 9 \end{vmatrix}$$

Since, the two columns C_2 and C_3

are identical.

$$\therefore \Delta = 0$$

$$\text{Hence, } \begin{vmatrix} 2 & 7 & 65 \\ 3 & 8 & 75 \\ 5 & 9 & 86 \end{vmatrix} = 0$$

4. Using the property of determinants and without expanding, prove that:

$$\begin{vmatrix} 1 & bc & a(b+c) \\ 1 & ca & b(c+a) \\ 1 & ab & c(a+b) \end{vmatrix} = 0.$$

$$\text{Ans: Let } \Delta = \begin{vmatrix} 1 & bc & a(b+c) \\ 1 & ca & b(c+a) \\ 1 & ab & c(a+b) \end{vmatrix}$$

Applying the column operation, $C_3 \rightarrow C_3 + C_2$.

$$\Delta = \begin{vmatrix} 1 & bc & ab+bc+ca \\ 1 & ca & ab+bc+ca \\ 1 & ab & ab+bc+ca \end{vmatrix}$$

Taking $(ab + bc + ca)$ common from the third column, we get:

$$\Delta = (ab + bc + ca) \begin{vmatrix} 1 & bc & 1 \\ 1 & ca & 1 \\ 1 & ab & 1 \end{vmatrix}$$

Since, the two columns C_1 and C_3 are identical.

$$\therefore \Delta = 0$$

$$\text{Hence, } \begin{vmatrix} 1 & bc & a(b+c) \\ 1 & ca & b(c+a) \\ 1 & ab & c(a+b) \end{vmatrix} = 0.$$

5. Using the property of determinants and without expanding, prove that:

$$\begin{vmatrix} b+c & q+r & y+z \\ c+a & r+p & z+x \\ a+b & p+q & x+y \end{vmatrix} = 2 \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix}.$$

$$\text{Ans: Let } \Delta = \begin{vmatrix} b+c & q+r & y+z \\ c+a & r+p & z+x \\ a+b & p+q & x+y \end{vmatrix}$$

Applying the Sum Property, we get

$$\Rightarrow \Delta = \begin{vmatrix} b+c & q+r & y+z \\ c+a & r+p & z+x \\ a & p & x \end{vmatrix} + \begin{vmatrix} b+c & q+r & y+z \\ c+a & r+p & z+x \\ b & q & y \end{vmatrix}$$

$$\text{Suppose } \Delta = \Delta_1 + \Delta_2 \dots\dots(1)$$

$$\text{Now, } \Delta_1 = \begin{vmatrix} b+c & q+r & y+z \\ c+a & r+p & z+x \\ a & p & x \end{vmatrix}$$

Applying the row operation, $R_2 \rightarrow R_2 - R_3$

$$\Rightarrow \Delta_1 = \begin{vmatrix} b+c & q+r & y+z \\ c & r & z \\ a & p & x \end{vmatrix}$$

Again, applying the row operation, $R_1 \rightarrow R_1 - R_2$

$$\Rightarrow \Delta_1 = \begin{vmatrix} b & q & y \\ c & r & z \\ a & p & x \end{vmatrix}$$

We know, that if any two rows or columns of a determinant are interchanged, the value of the determinant is multiplied by (-1) .

Hence, interchanging the rows, $R_1 \leftrightarrow R_2$ and $R_2 \leftrightarrow R_3$, we have

$$\Rightarrow \Delta_1 = (-1)^2 \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix}$$

$$\therefore \Delta_1 = \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix} \dots\dots(2)$$

$$\text{We have, } \Delta_1 = \begin{vmatrix} b+c & q+r & y+z \\ c+a & r+p & z+x \\ b & q & y \end{vmatrix}$$

Applying the row operation, $R_1 \rightarrow R_1 - R_3$

$$\Rightarrow \Delta_2 = \begin{vmatrix} c & r & z \\ c+a & r+p & z+x \\ b & q & y \end{vmatrix}$$

Applying the row operation, $R_2 \rightarrow R_2 - R_1$

$$\Rightarrow \Delta_2 = \begin{vmatrix} c & r & z \\ a & p & x \\ b & q & y \end{vmatrix}$$

Interchanging the rows, $R_1 \leftrightarrow R_2$ and $R_2 \leftrightarrow R_3$

$$\Rightarrow \Delta_2 = (-1)^2 \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix}$$

$$\therefore \Delta_2 = \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix} \dots\dots (3)$$

From (2) and (3), we get:

$$\Rightarrow \Delta_1 = \Delta_2 = \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix}$$

From (1), we have:

$$\Rightarrow \Delta = 2\Delta_1$$

$$\therefore \Delta = 2 \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix}.$$

$$\text{Hence, } \begin{vmatrix} b+c & q+r & y+z \\ c+a & r+p & z+x \\ a+b & p+q & x+y \end{vmatrix} = 2 \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix}.$$

6. By using properties of determinants, show that:

$$\begin{vmatrix} 0 & a & -b \\ -a & 0 & -c \\ b & c & 0 \end{vmatrix} = 0.$$

Ans: Given, $\Delta = \begin{vmatrix} 0 & a & -b \\ -a & 0 & -c \\ b & c & 0 \end{vmatrix}$

We know, that if we multiply the elements of a matrix by a scalar c , then we will multiply the matrix by the scalar, $\frac{1}{c}$.

Applying $R_1 \rightarrow cR_1$:

$$\Rightarrow \Delta = \frac{1}{c} \begin{vmatrix} 0 & ac & -bc \\ -a & 0 & -c \\ b & c & 0 \end{vmatrix}$$

Applying $R_1 \rightarrow R_1 - bR_2$

$$\Delta = \frac{1}{c} \begin{vmatrix} ab & ac & 0 \\ -a & 0 & -c \\ b & c & 0 \end{vmatrix}$$

Taking a common from the first row, we have:

$$\Rightarrow \Delta = \frac{a}{c} \begin{vmatrix} b & c & 0 \\ -a & 0 & -c \\ b & c & 0 \end{vmatrix}$$

Since, the two rows R_1 and R_3 are identical.

$$\therefore \Delta = 0$$

Hence,
$$\begin{vmatrix} 0 & a & -b \\ -a & 0 & -c \\ b & c & 0 \end{vmatrix} = 0.$$

7. By using properties of determinants, show that:

$$\begin{vmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ca & cb & -c^2 \end{vmatrix} = 4a^2b^2c^2.$$

Ans: Let $\Delta = \begin{vmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ca & cb & -c^2 \end{vmatrix}$

Taking out a,b,c from R_1 , R_2 and R_3 respectively, we have:

$$\Rightarrow \Delta = abc \begin{vmatrix} -a & b & c \\ a & -b & c \\ a & b & -c \end{vmatrix}$$

Similarly, taking out a,b,c from C_1 , C_2 and C_3 respectively, we have:

$$\Rightarrow \Delta = a^2b^2c^2 \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

Applying the row operations $R_2 \rightarrow R_2 + R_1$ and $R_3 \rightarrow R_3 + R_1$

$$\Rightarrow \Delta = a^2 b^2 c^2 \begin{vmatrix} -1 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \end{vmatrix}$$

Solving it along the first column, C_1 we get:

$$\Rightarrow \Delta = a^2 b^2 c^2 (-1) \begin{vmatrix} 0 & 2 \\ 2 & 0 \end{vmatrix}$$

$$\Rightarrow \Delta = a^2 b^2 c^2 (0-4)$$

$$\therefore \Delta = 4a^2 b^2 c^2$$

$$\text{Hence, } \begin{vmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ca & cb & -c^2 \end{vmatrix} = 4a^2 b^2 c^2.$$

8. By using properties of determinants, show that:

$$\text{i. } \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$$

$$\text{Ans: Let } \Delta = \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

Applying the row operations $R_1 \rightarrow R_1 - R_3$ and $R_2 \rightarrow R_2 - R_3$

$$\Rightarrow \Delta = \begin{vmatrix} 0 & a-c & a^2-c^2 \\ 0 & b-c & b^2-c^2 \\ 1 & c & c^2 \end{vmatrix}$$

We know, $a^2 - b^2 = (a + b)(a - b)$

Thus,

$$\Rightarrow \Delta = (c - a)(b - c) \begin{vmatrix} 0 & -1 & -(a + c) \\ 0 & 1 & b + c \\ 1 & c & c^2 \end{vmatrix}$$

Applying $R_1 \rightarrow R_1 + R_2$

$$\Rightarrow \Delta = (b - c)(c - a) \begin{vmatrix} 0 & 0 & -a + b \\ 0 & 1 & b + c \\ 1 & c & c^2 \end{vmatrix}$$

$$\Rightarrow \Delta = (b - c)(c - a) \begin{vmatrix} 0 & 0 & -(a - b) \\ 0 & 1 & b + c \\ 1 & c & c^2 \end{vmatrix}$$

Taking out $(a - b)$ common from R_1

$$\Rightarrow \Delta = (a - b)(b - c)(c - a) \begin{vmatrix} 0 & 0 & -1 \\ 0 & 1 & b + c \\ 1 & c & c^2 \end{vmatrix}$$

Expanding along C_1 ,

$$\Rightarrow \Delta = (a - b)(b - c)(c - a) \begin{vmatrix} 0 & 1 \\ 1 & b + c \end{vmatrix}$$

$$\therefore \Delta = (a - b)(b - c)(c - a)$$

$$\text{Hence, } \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a - b)(b - c)(c - a)$$

$$\text{ii. } \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$$

$$\text{Ans: Let } \Delta = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix}$$

Applying the column operations, $C_1 \rightarrow C_1 - C_3$ and $C_2 \rightarrow C_2 - C_3$

$$\Rightarrow \Delta = \begin{vmatrix} 0 & 0 & 1 \\ a-c & b-c & c \\ a^3-c^3 & b^3-c^3 & c^3 \end{vmatrix}$$

We know, $x^3 - y^3 = (x-y)(x^2 + y^2 + xy)$.

$$\Rightarrow \Delta = \begin{vmatrix} 0 & 0 & 1 \\ a-c & b-c & c \\ (a-c)(a^2+ac+c^2) & (b-c)(b^2+bc+c^2) & c^3 \end{vmatrix}$$

Taking out $(c-a)$ and $(b-c)$ common from C_1 and C_2 respectively,

$$\Rightarrow \Delta = (c-a)(b-c) \begin{vmatrix} 0 & 0 & 1 \\ -1 & 1 & c \\ -(a^2+ac+c^2) & (b^2+bc+c^2) & c^3 \end{vmatrix}$$

Applying $C_1 \rightarrow C_1 + C_2$

$$\Rightarrow \Delta = (c-a)(b-c) \begin{vmatrix} 0 & 0 & 1 \\ 0 & 1 & c \\ (b^2-a^2)+c(b-a) & (b^2+bc+c^2) & c^3 \end{vmatrix}$$

Taking out $(a - c)$ common from C_1 , we get:

$$\Rightarrow \Delta = (a - b)(c - a)(b - c) \begin{vmatrix} 0 & 0 & 1 \\ 0 & 0 & c \\ -(a+b+c) & (b^2+bc+c^2) & c^3 \end{vmatrix}$$

Again taking out $(a + b + c)$ common from C_1 , we get:

$$\Rightarrow \Delta = (a - b)(b - c)(c - a)(a + b + c) \begin{vmatrix} 0 & 0 & 1 \\ 0 & 0 & c \\ -1 & (b^2+bc+c^2) & c^3 \end{vmatrix}$$

Expanding along C_1 , we get:

$$\Rightarrow \Delta = (a - b)(b - c)(c - a)(a + b + c)(-1) \begin{vmatrix} 0 & 1 \\ 1 & c \end{vmatrix}$$

$$\therefore \Delta = (a - b)(b - c)(c - a)(a + b + c)$$

$$\text{Hence, } \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a - b)(b - c)(c - a)(a + b + c)$$

9. By using properties of determinants, show that:

$$\begin{vmatrix} x & x^2 & yz \\ y & y^2 & zx \\ z & z^2 & xy \end{vmatrix} = (x - y)(y - z)(z - x)(xy + yz + zx).$$

Ans: Let $\Delta = \begin{vmatrix} x & x^2 & yz \\ y & y^2 & zx \\ z & z^2 & xy \end{vmatrix}$

Applying the row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$

$$\Rightarrow \Delta = \begin{vmatrix} x & x^2 & yz \\ y-x & y^2-x^2 & zx-yz \\ z-x & z^2-x^2 & xy-yz \end{vmatrix}$$

$$\Rightarrow \Delta = \begin{vmatrix} x & x^2 & yz \\ -(x-y) & -(x-y)(x+y) & z(x-y) \\ (z-x) & (z-x)(z+x) & -y(z-x) \end{vmatrix}$$

Taking out $(x-y)$ and $(z-x)$ common from R_2 and R_3 respectively

$$\Rightarrow \Delta = (x-y)(z-x) \begin{vmatrix} x & x^2 & yz \\ -1 & -x-y & z \\ 1 & z+x & -y \end{vmatrix}$$

Applying $R_3 \rightarrow R_3 + R_2$

$$\Rightarrow \Delta = (x-y)(z-x) \begin{vmatrix} x & x^2 & yz \\ -1 & -x-y & z \\ 0 & z-y & z-y \end{vmatrix}$$

Taking out $(z-y)$ common from R_3 , we get:

$$\Rightarrow \Delta = (x-y)(z-x)(z-y) \begin{vmatrix} x & x^2 & yz \\ -1 & -x-y & z \\ 0 & 1 & 1 \end{vmatrix}$$

Expanding along R_3

$$\Rightarrow \Delta = [(x-y)(z-x)(z-y)] \left[(-1) \begin{vmatrix} x & yz \\ -1 & z \end{vmatrix} + 1 \begin{vmatrix} x & x^2 \\ -1 & -x-y \end{vmatrix} \right]$$

$$\Rightarrow \Delta = (x-y)(z-x)(z-y) [(-xz-yz) + (-x^2-xy+x^2)]$$

$$\Rightarrow \Delta = -(x-y)(z-x)(z-y)(xy+yz+zx)$$

$$\therefore \Delta = (x-y)(y-z)(z-x)(xy+yz+zx)$$

$$\text{Hence, } \begin{vmatrix} x & x^2 & yz \\ y & y^2 & zx \\ z & z^2 & xy \end{vmatrix} = (x-y)(y-z)(z-x)(xy+yz+zx)$$

10. By using properties of determinants, show that:

$$\text{i. } \begin{vmatrix} x+4 & 2x & 2x \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix} = (5x+4)(4-x)^2$$

$$\text{Ans: Let } \Delta = \begin{vmatrix} x+4 & 2x & 2x \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix}$$

Applying the row operation, $R_1 \rightarrow R_1 + R_2 + R_3$

$$\Rightarrow \Delta = \begin{vmatrix} 5x+4 & 5x+4 & 5x+4 \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix}$$

Taking out $(5x+4)$ common from R_1

$$\Rightarrow \Delta = (5x+4) \begin{vmatrix} 1 & 1 & 1 \\ 2x & x+4 & 2x \\ 2x & 0 & x+4 \end{vmatrix}$$

Applying the column operations, $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$

$$\Rightarrow \Delta = (5x+4) \begin{vmatrix} 1 & 0 & 0 \\ 2x & 4-x & 0 \\ 2x & 0 & 4-x \end{vmatrix}$$

Taking out $(4-x)$ common from C_2 and C_3 respectively,

$$\Rightarrow \Delta = (5x+4)(4-x)(4-x) \begin{vmatrix} 1 & 0 & 0 \\ 2x & 1 & 0 \\ 2x & 0 & 1 \end{vmatrix}$$

Expanding along C_3

$$\Rightarrow \Delta = (5x+4)(4-x)^2 \begin{vmatrix} 1 & 0 \\ 2x & 1 \end{vmatrix}$$

$$\therefore \Delta = (5x+4)(4-x)^2$$

$$\text{Hence, } \begin{vmatrix} x+4 & 2x & 2x \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix} = (5x+4)(4-x)^2$$

$$\text{ii. } \begin{vmatrix} y+k & y & y \\ y & y+k & y \\ y & y & y+k \end{vmatrix} = k^2(3x+k)$$

Ans: Let $\Delta = \begin{vmatrix} y+k & y & y \\ y & y+k & y \\ y & y & y+k \end{vmatrix}$

Applying the row operation, $R_1 \rightarrow R_1 + R_2 + R_3$

$$\Rightarrow \Delta = \begin{vmatrix} 3y+k & 3y+k & 3y+k \\ y & y+k & y \\ y & y & y+k \end{vmatrix}$$

Taking out $(3y+k)$ common from R_1

$$\Rightarrow \Delta = (3y+k) \begin{vmatrix} 1 & 1 & 1 \\ y & y+k & y \\ y & y & y+k \end{vmatrix}$$

Applying the column operations, $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$

$$\Rightarrow \Delta = (3y+k) \begin{vmatrix} 1 & 0 & 0 \\ y & k & 0 \\ y & 0 & k \end{vmatrix}$$

Taking out (k) common from C_2 and C_3 respectively,

$$\Rightarrow \Delta = k^2 (3y+k) \begin{vmatrix} 1 & 0 & 0 \\ y & 1 & 0 \\ y & 0 & 1 \end{vmatrix}$$

Expanding along C_3

$$\Rightarrow \Delta = k^2 (3y+k) \begin{vmatrix} 1 & 0 \\ y & 1 \end{vmatrix}$$

$$\therefore \Delta = k^2 (3y+k)$$

Hence, $\begin{vmatrix} y+k & y & y \\ y & y+k & y \\ y & y & y+k \end{vmatrix} = k^2(3x+k).$

11. By using properties of determinants, show that:

i. $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3$

Ans: Let $\Delta = \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$

Applying the row operation, $R_1 \rightarrow R_1 + R_2 + R_3$

$$\Rightarrow \Delta = \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

Taking out $(a+b+c)$ common from R_1

$$\Rightarrow \Delta = (a+b+c) \begin{vmatrix} 1 & 1 & 1 \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

Applying the column operations, $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$

$$\Rightarrow \Delta = (a+b+c) \begin{vmatrix} 1 & 0 & 0 \\ 2b & -(a+b+c) & 0 \\ 2c & 0 & -(a+b+c) \end{vmatrix}$$

Taking out $(a+b+c)$ common from C_2 and C_3 respectively,

$$\Rightarrow \Delta = (a+b+c)^3 \begin{vmatrix} 1 & 0 & 0 \\ 2b & -1 & 0 \\ 2c & 0 & -1 \end{vmatrix}$$

Expanding along C_3

$$\Rightarrow \Delta = (a+b+c)^3 (-1)(-1)$$

$$\therefore \Delta = (a+b+c)^3$$

Hence, $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3.$

ii. $\begin{vmatrix} x+y+2z & x & y \\ z & y+z+2z & y \\ z & x & z+x+2y \end{vmatrix} = 2(x+y+z)^3$

Ans: Let $\Delta = \begin{vmatrix} x+y+2z & x & y \\ z & y+z+2z & y \\ z & x & z+x+2y \end{vmatrix}$

Applying the column operation, $C_1 \rightarrow C_1 + C_2 + C_3$

$$\Rightarrow \Delta = \begin{vmatrix} 2(x+y+z) & x & y \\ 2(x+y+z) & y+z+2z & y \\ 2(x+y+z) & x & z+x+2y \end{vmatrix}$$

Taking out $2(x+y+z)$ common from C_1

$$\Rightarrow \Delta = 2(x+y+z) \begin{vmatrix} 1 & x & y \\ 1 & y+z+2z & y \\ 1 & x & z+x+2y \end{vmatrix}$$

Applying the row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$

$$\Rightarrow \Delta = 2(x+y+z) \begin{vmatrix} 1 & x & y \\ 0 & x+y+z & 0 \\ 0 & 0 & x+y+z \end{vmatrix}$$

Taking out $(x+y+z)$ common from R_2 and R_3 respectively,

$$\Rightarrow \Delta = 2(x+y+z) \begin{vmatrix} 1 & x & y \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

Expanding along R_3

$$\Rightarrow \Delta = 2(x+y+z)^3 (1)(1-0)$$

$$\therefore \Delta = 2(x+y+z)^3$$

Hence, $\begin{vmatrix} x+y+2z & x & y \\ z & y+z+2z & y \\ z & x & z+x+2y \end{vmatrix} = 2(x+y+z)^3.$

12. By using properties of determinants, show that:

$$\begin{vmatrix} 1 & x & x^2 \\ x^2 & 1 & x \\ x & x^2 & 1 \end{vmatrix} = (1-x^3)^2.$$

Ans: Let $\Delta = \begin{vmatrix} 1 & x & x^2 \\ x^2 & 1 & x \\ x & x^2 & 1 \end{vmatrix}$

Applying $R_1 \rightarrow R_1 + R_2 + R_3$

$$\Rightarrow \Delta = \begin{vmatrix} 1+x+x^2 & 1+x+x^2 & 1+x+x^2 \\ x^2 & 1 & x \\ x & x^2 & 1 \end{vmatrix}$$

Taking out $(1+x+x^2)$ common from R_1

$$\Rightarrow \Delta = (1+x+x^2) \begin{vmatrix} 1 & 1 & 1 \\ x^2 & 1 & x \\ x & x^2 & 1 \end{vmatrix}$$

Applying $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$

$$\Rightarrow \Delta = (1+x+x^2) \begin{vmatrix} 1 & 0 & 0 \\ x^2 & 1-x^2 & x-x^2 \\ x & x^2-x & 1-x \end{vmatrix}$$

Taking out $(1-x)$ common from C_2 and C_3 respectively,

$$\Rightarrow \Delta = (1+x+x^2)(1-x)(1-x) \begin{vmatrix} 1 & 0 & 0 \\ x^2 & 1+x & x \\ x & -x & 1 \end{vmatrix}$$

$$\Rightarrow \Delta = (1-x^3)(1-x) \begin{vmatrix} 1 & 0 & 0 \\ x^2 & 1+x & x \\ x & -x & 1 \end{vmatrix}$$

Expanding along R_1 .

$$\Rightarrow \Delta = (1-x^3)(1-x)(1) \begin{vmatrix} 1+x & x \\ -x & 1 \end{vmatrix}$$

$$\Rightarrow \Delta = (1-x^3)(1-x)(1+x+x^2)$$

$$\Rightarrow \Delta = (1-x^3)(1-x^3)$$

$$\therefore \Delta = (1-x^3)^2$$

Hence, $\begin{vmatrix} 1 & x & x^2 \\ x^2 & 1 & x \\ x & x^2 & 1 \end{vmatrix} = (1-x^3)^2$.

13. By using properties of determinants, show that:

$$\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} = (1+a^2+b^2)^3$$

Ans: Let $\Delta = \begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix}$

Applying the row operations, $R_1 \rightarrow R_1 + bR_3$ and $R_2 \rightarrow R_2 - R_3$

$$\Rightarrow \Delta = \begin{vmatrix} 1+a^2+b^2 & 0 & -b(1+a^2+b^2) \\ 0 & 1+a^2+b^2 & a(1+a^2+b^2) \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix}$$

Taking out $(1+a^2+b^2)$ common from R_1 and R_2 respectively,

$$\Rightarrow \Delta = (1+a^2+b^2)^2 \begin{vmatrix} 1 & 0 & -b \\ 0 & 1 & a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix}$$

Expanding along R_1

$$\Rightarrow \Delta = (1+a^2+b^2)^2 \left[(1) \begin{vmatrix} 1 & a \\ -2a & 1-a^2-b^2 \end{vmatrix} - b \begin{vmatrix} 0 & 1 \\ 2b & -2a \end{vmatrix} \right]$$

$$\Rightarrow \Delta = (1+a^2+b^2)^2 [1 - 1 - a^2 - b^2 + 2a^2 - b(-2b)]$$

$$\Rightarrow \Delta = (1+a^2+b^2)^2 (1+a^2+b^2)$$

$$\therefore \Delta = (1+a^2+b^2)^3$$

$$\text{Hence, } \begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} = (1+a^2+b^2)^3.$$

14. By using properties of determinants, show that:

$$\begin{vmatrix} a^2+1 & ab & ac \\ ab & b^2+1 & bc \\ ca & cb & c^2+1 \end{vmatrix} = 1+a^2+b^2+c^2.$$

$$\text{Ans: Let } \Delta = \begin{vmatrix} a^2+1 & ab & ac \\ ab & b^2+1 & bc \\ ca & cb & c^2+1 \end{vmatrix}$$

Taking out a, b and c from R_1 , R_2 and R_3 respectively

$$\Rightarrow \Delta = abc \begin{vmatrix} a + \frac{1}{a} & b & c \\ a & b + \frac{1}{b} & c \\ a & b & c + \frac{1}{c} \end{vmatrix}$$

Applying the row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$

$$\Rightarrow \Delta = abc \begin{vmatrix} a + \frac{1}{a} & b & c \\ -\frac{1}{a} & \frac{1}{b} & 0 \\ -\frac{1}{a} & 0 & \frac{1}{c} \end{vmatrix}$$

Applying $C_1 \rightarrow aC_1$, $C_2 \rightarrow bC_2$ and $C_3 \rightarrow cC_3$

$$\Rightarrow \Delta = abc \times \frac{1}{abc} \begin{vmatrix} a^2 + 1 & b^2 & c^2 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix}$$

Expanding along C_3

$$\Rightarrow \Delta = -1 \begin{vmatrix} b^2 & c^2 \\ 1 & 0 \end{vmatrix} + 1 \begin{vmatrix} a^2 + 1 & b^2 \\ -1 & 1 \end{vmatrix}$$

$$\Rightarrow \Delta = 1(-c^2) + (a^2 + 1 + b^2)$$

$$\therefore \Delta = 1 + a^2 + b^2 + c^2$$

$$\text{Hence, } \begin{vmatrix} a^2 + 1 & ab & ac \\ ab & b^2 + 1 & bc \\ ca & cb & c^2 + 1 \end{vmatrix} = 1 + a^2 + b^2 + c^2.$$

15. Choose the correct answer. Let A be a square matrix of order 3×3 , then

A. $|kA|$ is equal to $k|A|$

B. $k^2|A|$

C. $k^3|A|$

D. $3k|A|$

Ans: Since, A is a square matrix of order 3×3 .

$$\text{Let us suppose } A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

$$\text{Thus, } kA = \begin{bmatrix} ka_1 & kb_1 & kc_1 \\ ka_2 & kb_2 & kc_2 \\ ka_3 & kb_3 & kc_3 \end{bmatrix}$$

$$\therefore |kA| = \begin{vmatrix} ka_1 & kb_1 & kc_1 \\ ka_2 & kb_2 & kc_2 \\ ka_3 & kb_3 & kc_3 \end{vmatrix}$$

Taking out (k) common from each row, we have:

$$\Rightarrow |kA| = k^3 \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$\therefore |kA| = k^3 |A|$$

Hence, **B. $k^3|A|$** is the correct option.

16. Which of the following is correct?

A. Determinant is a square matrix.

- B. Determinant is a number associated to a matrix.**
C. Determinant is a number associated to a square matrix.
D. None of these.

Ans: For every square matrix, $A = [a_{ij}]$ of order n , we can determine or associate a value which is termed as determinant of square matrix A , where $a_{ij} = (i,j)^{\text{th}}$ element of A .

Thus, the determinant is a number associated to a square matrix.

Hence, **C. Determinant is a number associated to a square matrix** is the correct option.

Exercise 4.3

1. Find area of the triangle with vertices at the point given in each of the following:

i. $(1,0), (6,0), (4,3)$

Ans: Given vertices, $(1,0), (6,0), (4,3)$

We know, if we have three points (x_1, y_1) , (x_2, y_2) and (x_3, y_3) , then the area of the triangle is given by,

$$\Delta = \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

Thus, the area of the triangle is given by,

$$\Rightarrow \Delta = \frac{1}{2} \begin{vmatrix} 1 & 0 & 1 \\ 6 & 0 & 1 \\ 4 & 3 & 1 \end{vmatrix}$$

$$\Rightarrow \Delta = \frac{1}{2} [1(0-3) - 0(6-4) + 1(18-0)]$$

$$\Rightarrow \Delta = \frac{1}{2} [-3 + 18]$$

$$\Rightarrow \Delta = \frac{15}{2} \text{ square units}$$

$\therefore$ Area of the triangle with vertices $(1,0), (6,0), (4,3)$ is $\frac{15}{2}$ square units.

ii. $(2,7), (1,1), (10,8)$

Ans: Given vertices, $(2,7), (1,1), (10,8)$

We know, if we have three points (x_1, y_1) , (x_2, y_2) and (x_3, y_3) , then the area of the triangle is given by,

$$\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

Thus, the area of the triangle is given by,

$$\Rightarrow \Delta = \frac{1}{2} \begin{vmatrix} 2 & 7 & 1 \\ 1 & 1 & 1 \\ 10 & 8 & 1 \end{vmatrix}$$

$$\Rightarrow \Delta = \frac{1}{2} [2(1-8) - 7(1-10) + 1(8-10)]$$

$$\Rightarrow \Delta = \frac{1}{2} [2(-7) - 7(-9) + 1(-2)]$$

$$\Rightarrow \Delta = \frac{1}{2}[-16+63]$$

$$\Rightarrow \Delta = \frac{47}{2} \text{ square units}$$

$\therefore$ Area of the triangle with vertices $(2,7), (1,1), (10,8)$ is $\frac{47}{2}$ square units.

iii. $(-2,-3), (3,2), (-1,-8)$

Ans: Given vertices, $(-2,-3), (3,2), (-1,-8)$

We know, if we have three points (x_1, y_1) , (x_2, y_2) and (x_3, y_3) , then the area of the triangle is given by,

$$\Delta = \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

Thus, the area of the triangle with vertices $(-2,-3), (3,2), (-1,-8)$ is given by,

$$\Rightarrow \Delta = \frac{1}{2} \begin{vmatrix} -2 & -3 & 1 \\ 3 & 2 & 1 \\ -1 & -8 & 1 \end{vmatrix}$$

$$\Rightarrow \Delta = \frac{1}{2}[-2(2+8)+3(3+1)+1(-24+2)]$$

$$\Rightarrow \Delta = \frac{1}{2}[-20+12-22]$$

$$\Rightarrow \Delta = -\frac{30}{2}$$

$$\Rightarrow \Delta = -15$$

∴ The area of the triangle with vertices (2,7), (1,1), (10,8) is $|-15|=15$ square units.

2. Show that points A(a,b+c), B(b,c+a), C(c,a+b) are collinear.

Ans: To show that the points A(a,b+c), B(b,c+a), C(c,a+b) are collinear, the area of the triangle formed by these points as vertices should be zero.

∴ Area of ΔABC is given by,

$$\Rightarrow \Delta = \frac{1}{2} \begin{vmatrix} a & b+c & 1 \\ b & c+a & 1 \\ c & a+b & 1 \end{vmatrix}$$

Applying the row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$

$$\Rightarrow \Delta = \frac{1}{2} \begin{vmatrix} a & b+c & 1 \\ b-a & a-b & 0 \\ c-a & a-c & 0 \end{vmatrix}$$

Taking out (a-b) and (c-a) common from R_2 and R_3 respectively,

$$\Rightarrow \Delta = \frac{1}{2} (a-b)(c-a) \begin{vmatrix} a & b+c & 1 \\ -1 & 1 & 0 \\ 1 & -1 & 0 \end{vmatrix}$$

Applying the row operation $R_3 \rightarrow R_3 + R_2$

$$\Rightarrow \Delta = \frac{1}{2} (a-b)(c-a) \begin{vmatrix} a & b+c & 1 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{vmatrix}$$

Since all the elements of the last row of the matrix are zero then the value of the determinant will be 0.

$$\therefore \Delta = 0$$

Thus, the area of the triangle formed by points A, B and C is zero.

Hence, the points A(a,b+c), B(b,c+a), C(c,a+b) are collinear.

3. Find values of k if area of triangle is 4 square units and vertices are

i. (k,0), (4,0), (0,2)

Ans: Given vertices are (k,0), (4,0), (0,2).

We know, if we have three points (x_1, y_1) , (x_2, y_2) and (x_3, y_3) , then the area of the triangle is given by,

$$\Delta = \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

Thus, the area of the triangle is given by,

$$\Rightarrow \Delta = \frac{1}{2} \begin{vmatrix} k & 0 & 1 \\ 4 & 0 & 1 \\ 0 & 2 & 1 \end{vmatrix}$$

$$\Rightarrow \Delta = \frac{1}{2} [k(0-2) - 0(4-0) + 1(8-0)]$$

$$\Rightarrow \Delta = \frac{1}{2} [-2k + 8]$$

$$\therefore \Delta = -k + 4$$

Since the area is given to be 4 square units, thus

$$-k + 4 = \pm 4$$

When $-k + 4 = -4$

$$\therefore k = 8.$$

When $-k + 4 = 4$

$$\therefore k = 0.$$

Hence, $k=0,8$.

ii. $(-2,0),(0,4),(0,k)$

Ans: Given vertices are $(-2,0),(0,4),(0,k)$.

We know, if we have three points (x_1, y_1) , (x_2, y_2) and (x_3, y_3) , then the area of the triangle is given by,

$$\Delta = \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

The area of the triangle is given by,

$$\Rightarrow \Delta = \frac{1}{2} \begin{vmatrix} -2 & 0 & 1 \\ 0 & 4 & 1 \\ 0 & k & 1 \end{vmatrix}$$

$$\Rightarrow \Delta = \frac{1}{2} [-2(4-k)]$$

$$\Rightarrow \Delta = k - 4$$

Since the area is given to be 4 square units, thus

$$k - 4 = \pm 4$$

When $k - 4 = -4$

$$\therefore k = 0.$$

When $k - 4 = 4$

$$\therefore k = 8.$$

Hence, $k = 0, 8$.

4. Determine the following:

i. Find equation of line joining (1,2) and (3,6) using determinants.

Ans: Let us assume a point, $P(x, y)$ on the line joining points $A(1,2)$ and $B(3,6)$

Then, the point A, B and P are collinear.

Thus, the area of triangle ABP will be zero.

$$\therefore \frac{1}{2} \begin{vmatrix} 1 & 2 & 1 \\ 3 & 6 & 1 \\ x & y & 1 \end{vmatrix} = 0$$

$$\Rightarrow \frac{1}{2} [1(6-y) - 2(3-x) + 1(3y-6x)] = 0$$

$$\Rightarrow 6-y-6+2x+3y-6x=0$$

$$\Rightarrow 2y-4x=0$$

$$\Rightarrow y=2x$$

$\therefore$ The equation of the line joining the given points is $y = 2x$.

ii. Find equation of line joining (3,1) and (9,3) using determinants.

Ans: Let us assume a point, $P(x, y)$ on the line joining points $A(3,1)$ and $B(9,3)$.

Then, the point A, B and P are collinear.

Thus, the area of the triangle ABP will be zero.

$$\therefore \frac{1}{2} \begin{vmatrix} 3 & 1 & 1 \\ 9 & 3 & 1 \\ x & y & 1 \end{vmatrix} = 0$$

$$\Rightarrow \frac{1}{2} [3(3-y) - 1(9-x) + 1(9y-3x)] = 0$$

$$\Rightarrow 9 - 3y - 9 + x + 9y - 3x = 0$$

$$\Rightarrow 6y - 2x = 0$$

$$\Rightarrow x - 3y = 0$$

$\therefore$ The equation of the line joining the given points is $x - 3y = 0$.

5. If the area of triangle is 35 square units with vertices (2,-6) , (5,4) and (k,4) . Then k is

- A. 12
- B. -2
- C. -12,-2
- D. 12,-2

Ans: Given vertices, (2,-6), (5,4) and (k,4)

We know, if we have three points (x_1, y_1) , (x_2, y_2) and (x_3, y_3) , then the area of the triangle is given by,

$$\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

The area of the triangle is given by,

$$\Rightarrow \Delta = \frac{1}{2} \begin{vmatrix} 2 & -6 & 1 \\ 5 & 4 & 1 \\ k & 4 & 1 \end{vmatrix}$$

$$\Rightarrow \Delta = \frac{1}{2} [2(4-4) + 6(5-k) + 1(20-4k)]$$

$$\Rightarrow \Delta = \frac{1}{2} [50 - 10k]$$

$$\Rightarrow \Delta = 25 - 5k$$

Given, the area of the triangle is 35 square units .

Thus, we have:

$$\Rightarrow 25 - 5k = \pm 35$$

$$\Rightarrow 5(5 - k) = \pm 35$$

$$\Rightarrow 5 - k = \pm 7.$$

$$\text{When } 5 - k = 7$$

$$\therefore k = -2.$$

$$\text{When } 5 - k = -7$$

$$\therefore k = 12.$$

Hence, $k = 12, -2$.

Thus, D. $12, -2$ is the correct option.

Exercise 4.4

1. Write Minors and Cofactors of the elements of following determinants:

i. $\begin{vmatrix} 2 & -4 \\ 0 & 3 \end{vmatrix}$

Ans: Given, $\begin{vmatrix} 2 & -4 \\ 0 & 3 \end{vmatrix}$

Minor of an element is termed as the determinant obtained by removing the row and the column in which that element is present.

Minor of element a_{ij} is denoted by M_{ij} , where

i and j denotes the row and the column of the determinant respectively.

$$\therefore M_{11} = 3$$

$$M_{12} = 0$$

$$M_{21} = -4$$

$$M_{22} = 2$$

Cofactor of an element is termed as the determinant obtained by removing the row and the column in which that element is present preceded by a negative or a positive sign based on the position of the element.

Thus,

$$\text{Cofactor of } a_{ij} \text{ is } A_{ij} = (-1)^{i+j} M_{ij}$$

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11}$$

$$\Rightarrow A_{11} = (-1)^2 (3)$$

$$\therefore A_{11} = 3$$

Similarly,

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12}$$

$$\Rightarrow A_{12} = (-1)^3 (0)$$

$$\therefore A_{12}=0$$

$$\Rightarrow A_{21}=(-1)^{2+1} M_{21}$$

$$\Rightarrow A_{21}=(-1)^3 (-4)$$

$$\therefore A_{21}=4$$

$$\Rightarrow A_{22}=(-1)^{2+2} M_{22}$$

$$\Rightarrow A_{22}=(-1)^4 (2)$$

$$\therefore A_{22}=2$$

ii. $\begin{vmatrix} a & c \\ b & d \end{vmatrix}$

Ans: Given, $\begin{vmatrix} a & c \\ b & d \end{vmatrix}$

Minor of element a_{ij} is denoted by M_{ij} .

$$\therefore M_{11}=d$$

$$M_{12}=b$$

$$M_{21}=c$$

$$M_{22}=a$$

Cofactor of a_{ij} is $A_{ij}=(-1)^{i+j} M_{ij}$

$$\Rightarrow A_{11}=(-1)^{1+1} M_{11}$$

$$\Rightarrow A_{11}=(-1)^2 (d)$$

$$\therefore A_{11} = d$$

Similarly,

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12}$$

$$\Rightarrow A_{12} = (-1)^3 (b)$$

$$\therefore A_{12} = -b$$

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21}$$

$$\Rightarrow A_{21} = (-1)^3 (c)$$

$$\therefore A_{21} = -c$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22}$$

$$\Rightarrow A_{22} = (-1)^4 (a)$$

$$\therefore A_{22} = a$$

2. Write Minors and Cofactors of the elements of following determinants:

i.
$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

Ans: Given determinant,
$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}.$$

Minor of element a_{ij} is denoted by M_{ij} .

$$\therefore M_{11} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

$$\Rightarrow M_{12} = \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix} = 0$$

$$\Rightarrow M_{13} = \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} = 0$$

$$\Rightarrow M_{21} = \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix} = 0$$

$$\Rightarrow M_{22} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

$$\Rightarrow M_{23} = \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} = 0$$

$$\Rightarrow M_{31} = \begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix} = 0$$

$$\Rightarrow M_{32} = \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} = 0$$

$$\Rightarrow M_{33} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

Cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$.

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = 1$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = 0$$

$$\Rightarrow A_{13} = (-1)^{1+3} M_{13} = 0$$

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = 0$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = 1$$

$$\Rightarrow A_{23} = (-1)^{2+3} M_{23} = 0$$

$$\Rightarrow A_{31} = (-1)^{3+1} M_{31} = 0$$

$$\Rightarrow A_{32} = (-1)^{3+2} M_{32} = 0$$

$$\Rightarrow A_{33} = (-1)^{3+3} M_{33} = 1$$

ii. $\begin{vmatrix} 1 & 0 & 4 \\ 3 & 5 & -1 \\ 0 & 1 & 2 \end{vmatrix}$

Ans: Given determinant, $\begin{vmatrix} 1 & 0 & 4 \\ 3 & 5 & -1 \\ 0 & 1 & 2 \end{vmatrix}$

Minor of element a_{ij} is denoted by M_{ij} .

$$\Rightarrow M_{11} = \begin{vmatrix} 5 & -1 \\ 1 & 2 \end{vmatrix} = 10 + 1 = 11$$

$$\Rightarrow M_{12} = \begin{vmatrix} 3 & -1 \\ 0 & 2 \end{vmatrix} = 6 - 0 = 6$$

$$\Rightarrow M_{13} = \begin{vmatrix} 3 & 5 \\ 0 & 1 \end{vmatrix} = 3 - 0 = 3$$

$$\Rightarrow M_{21} = \begin{vmatrix} 0 & 4 \\ 1 & 2 \end{vmatrix} = 0 - 4 = -4$$

$$\Rightarrow M_{22} = \begin{vmatrix} 1 & 4 \\ 0 & 2 \end{vmatrix} = 2 - 0 = 2$$

$$\Rightarrow M_{23} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 - 0 = 1$$

$$\Rightarrow M_{31} = \begin{vmatrix} 0 & 4 \\ 5 & -1 \end{vmatrix} = 0 - 20 = -20$$

$$\Rightarrow M_{32} = \begin{vmatrix} 1 & 4 \\ 3 & -1 \end{vmatrix} = -1 - 12 = -13$$

$$\Rightarrow M_{33} = \begin{vmatrix} 1 & 0 \\ 3 & 5 \end{vmatrix} = 5 - 0 = 5$$

Cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$.

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = 11$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = 6$$

$$\Rightarrow A_{13} = (-1)^{1+3} M_{13} = 3$$

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = -4$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = 2$$

$$\Rightarrow A_{23} = (-1)^{2+3} M_{23} = 1$$

$$\Rightarrow A_{31} = (-1)^{3+1} M_{31} = -20$$

$$\Rightarrow A_{32} = (-1)^{3+2} M_{32} = -13$$

$$\Rightarrow A_{33} = (-1)^{3+3} M_{33} = 5$$

3. Using Cofactors of elements of second row, evaluate $\Delta = \begin{vmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}$.

Ans: Given determinant, $\begin{vmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}$

Determining the minors and cofactors, we get:

$$\Rightarrow M_{21} = \begin{vmatrix} 3 & 8 \\ 2 & 3 \end{vmatrix} = 9 - 16 = -7$$

$$\therefore A_{21} = (-1)^{2+1} M_{21} = 7$$

$$\Rightarrow M_{22} = \begin{vmatrix} 5 & 8 \\ 1 & 3 \end{vmatrix} = 15 - 8 = 7$$

$$\therefore A_{22} = (-1)^{2+2} M_{22} = 7$$

$$\Rightarrow M_{23} = \begin{vmatrix} 5 & 3 \\ 1 & 2 \end{vmatrix} = 10 - 3 = 7$$

$$\therefore A_{23} = (-1)^{2+1} M_{23} = -7$$

Since, Δ is equal to the sum of the product of the elements of the second row with their corresponding cofactors.

$$\therefore \Delta = a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23}$$

$$\Rightarrow \Delta = 2(7) + 0(7) + 1(7)$$

Hence, $\Delta = 21$.

4. Using Cofactors of elements of third column, evaluate $\Delta = \begin{vmatrix} 1 & x & yz \\ 1 & y & zx \\ 1 & z & xy \end{vmatrix}$.

Ans: Given determinant, $\begin{vmatrix} 1 & x & yz \\ 1 & y & zx \\ 1 & z & xy \end{vmatrix}$

Determining the minors and cofactors, we get:

$$\Rightarrow M_{13} = \begin{vmatrix} 1 & y \\ 1 & z \end{vmatrix} = z - y$$

$$\therefore A_{13} = (-1)^{1+3} M_{13} = (z - y)$$

$$\Rightarrow M_{23} = \begin{vmatrix} 1 & x \\ 1 & z \end{vmatrix} = z - x$$

$$\therefore A_{23} = (-1)^{2+3} M_{23} = (x - z)$$

$$\Rightarrow M_{33} = \begin{vmatrix} 1 & x \\ 1 & y \end{vmatrix} = y - x$$

$$\therefore A_{33} = (-1)^{3+3} M_{33} = (y - x)$$

Since, Δ is equal to the sum of the product of the elements of the first row with their corresponding cofactors.

$$\therefore \Delta = a_{13}A_{13} + a_{23}A_{23} + a_{33}A_{33}$$

$$\Rightarrow \Delta = yz(z - y) + zx(x - z) + xy(y - x)$$

$$\Rightarrow \Delta = yz^2 - y^2z + x^2z - xz^2 + xy^2 - x^2y$$

$$\Rightarrow \Delta = (x^2z - y^2z) + (yz^2 - xz^2) + (xy^2 - x^2y)$$

$$\Rightarrow \Delta = (x - y)[zx + zy - z^2 - xy]$$

$$\Rightarrow \Delta = (x - y)[z(x - z) + y(z - x)]$$

$$\text{Thus, } \Delta = (x - y)(y - z)(z - x).$$

5. If $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ and A_{ij} is Cofactors of a_{ij} , then value of Δ is given by

a) $a_{11}A_{31} + a_{12}A_{32} + a_{13}A_{33}$

b) $a_{11}A_{11} + a_{12}A_{21} + a_{13}A_{31}$

c) $a_{21}A_{11} + a_{22}A_{12} + a_{23}A_{13}$

d) $a_{11}A_{11} + a_{21}A_{21} + a_{31}A_{31}$

Ans: It is given that $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}.$

The value of Δ by expanding along first column is obtained as,

$$a_{11}(a_{22} \cdot a_{33} - a_{23} \cdot a_{32}) - a_{21}(a_{12} \cdot a_{33} - a_{13} \cdot a_{32}) + a_{31}(a_{12} \cdot a_{23} - a_{13} \cdot a_{22}) \dots 1$$

Now, the cofactor A_{ij} of element a_{ij} is given by $(-1)^{i+j} M_{ij}$, where M_{ij} is the minor. Minor is the determinant obtained by cancelling the i th row and j th column of the original matrix.

Now for element a_{11} , the minor is $M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = a_{22} \cdot a_{33} - a_{23} \cdot a_{32}$ and the cofactor is $A_{11} = (-1)^{1+1} (a_{22} \cdot a_{33} - a_{23} \cdot a_{32}) \Rightarrow A_{11} = (a_{22} \cdot a_{33} - a_{23} \cdot a_{32})$.

Next for element a_{21} , the minor is $M_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} = a_{12} \cdot a_{33} - a_{13} \cdot a_{32}$ and the cofactor is $A_{21} = (-1)^{2+1} (a_{12} \cdot a_{33} - a_{13} \cdot a_{32}) \Rightarrow A_{21} = -(a_{12} \cdot a_{33} - a_{13} \cdot a_{32})$.

Next for element a_{31} , the minor is $M_{31} = \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} = a_{12} \cdot a_{23} - a_{13} \cdot a_{22}$ and the cofactor is $A_{31} = (-1)^{3+1} (a_{12} \cdot a_{23} - a_{13} \cdot a_{22}) \Rightarrow A_{31} = (a_{12} \cdot a_{23} - a_{13} \cdot a_{22})$.

Now substituting the terms as obtained from above computation in equation 1,

$$a_{11}A_{11} - a_{21}(-A_{21}) + a_{31}A_{31}$$

$$a_{11}A_{11} + a_{21}A_{21} + a_{31}A_{31}$$

This matches with option d.

Exercise 4.5

1. Find the adjoint of each of the matrices. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

Ans: Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

Since, Cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$.

Thus,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = 4$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = -3$$

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = -2$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = 1$$

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\text{Thus, } \text{adj}A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}^T$$

$$\therefore \text{adj}A = \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}.$$

2. Find adjoint of each of the matrices $\begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \\ -2 & 0 & 1 \end{bmatrix}.$

$$\text{Ans: Let } A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \\ -2 & 0 & 1 \end{bmatrix}$$

Since, Cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$.

Thus,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = \begin{vmatrix} 3 & 5 \\ 0 & 1 \end{vmatrix} = 3 - 0 = 3$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = \begin{vmatrix} 2 & 5 \\ -2 & 1 \end{vmatrix} = -(2+10) = -12$$

$$\Rightarrow A_{13} = (-1)^{1+3} M_{13} = (-1)^4 M_{13}$$

$$\Rightarrow A_{13} = \begin{vmatrix} 2 & 3 \\ -2 & 0 \end{vmatrix} = 0+6=6$$

Similarly,

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = \begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} = -(-1-0)=1$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = \begin{vmatrix} 1 & 2 \\ -2 & 1 \end{vmatrix} = 1+4=5$$

$$\Rightarrow A_{23} = (-1)^{2+3} M_{23} = (-1)^5 M_{23}$$

$$\Rightarrow A_{23} = \begin{vmatrix} 1 & -1 \\ -2 & 0 \end{vmatrix} = (0-2)=-2$$

and

$$\Rightarrow A_{31} = (-1)^{3+1} M_{31} = (-1)^4 M_{31}$$

$$\Rightarrow A_{31} = \begin{vmatrix} -1 & 2 \\ 2 & 5 \end{vmatrix} = -5 - 4 = -9$$

$$\Rightarrow A_{32} = (-1)^{3+2} M_{32} = (-1)^5 M_{32}$$

$$\Rightarrow A_{32} = \begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix} = -(5 - 4) = -1$$

$$\Rightarrow A_{33} = (-1)^{3+3} M_{33} = (-1)^6 M_{33}$$

$$\Rightarrow A_{33} = \begin{vmatrix} 1 & -1 \\ 2 & 3 \end{vmatrix} = 3 + 2 = 5.$$

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\text{Thus, } \text{adj}A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^T = \begin{bmatrix} 3 & -12 & 6 \\ 1 & 5 & 2 \\ -9 & -1 & 5 \end{bmatrix}$$

$$3. \text{ Verify } A(\text{adj}A) = (\text{adj}A)A = |A|I. \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix}$$

$$\text{Ans: Given, } A = \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix}$$

$$\therefore |A| = -12 - (-12)$$

$$\Rightarrow |A| = 0$$

$$\text{Hence, } |A|I = 0 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow |A|I = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Since, Cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$.

Then,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = -6$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = 4$$

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = -3$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = 2$$

Cofactor matrix is $\begin{bmatrix} -6 & 4 \\ -3 & 2 \end{bmatrix}$.

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\text{Thus, } \text{adj} A = \begin{bmatrix} -6 & -3 \\ 4 & 2 \end{bmatrix}$$

Now, multiplying A with its adjoint, we have:

$$\Rightarrow A(\text{adj} A) = \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix} \begin{bmatrix} -6 & -3 \\ 4 & 2 \end{bmatrix}$$

$$\Rightarrow A(\text{adj}A) = \begin{bmatrix} -12+12 & -6+6 \\ 24-24 & 12-12 \end{bmatrix}$$

$$\therefore A(\text{adj}A) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Similarly, multiplying $(\text{adj}A)$ with A , we get:

$$\Rightarrow (\text{adj}A)A = \begin{bmatrix} -6 & -3 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix}$$

$$\Rightarrow (\text{adj}A)A = \begin{bmatrix} -12+12 & -18+18 \\ 8-8 & 12-12 \end{bmatrix}$$

$$\therefore (\text{adj}A)A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Thus, $A(\text{adj}A) = (\text{adj}A)A = |A|I$

Hence verified.

4. Verify $A(\text{adj}A) = (\text{adj}A)A = |A|I$. $\begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{bmatrix}$.

Ans: Let $A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{bmatrix}$

$$\Rightarrow |A| = 1(0-0) + 1(9+2) + 2(0-0)$$

$$\therefore |A| = 11$$

$$|A|_{I=11} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix}$$

Since, Cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$.

Thus,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = 0$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = -(9 + 2) = -11$$

$$\Rightarrow A_{13} = (-1)^{1+3} M_{13} = (-1)^4 M_{13}$$

$$\Rightarrow A_{13} = 0$$

Similarly,

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = -(-3 + 0) = 3$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = 3 - 2 = 1$$

$$\Rightarrow A_{23} = (-1)^{2+3} M_{23} = (-1)^5 M_{23}$$

$$\Rightarrow A_{23} = -(0 + 1) = -1$$

and

$$\Rightarrow A_{31} = (-1)^{3+1} M_{31} = (-1)^4 M_{31}$$

$$\Rightarrow A_{31} = 2 - 0 = 2$$

$$\Rightarrow A_{32} = (-1)^{3+2} M_{32} = (-1)^5 M_{32}$$

$$\Rightarrow A_{32} = -(-2 - 6) = 8$$

$$\Rightarrow A_{33} = (-1)^{3+3} M_{33} = (-1)^6 M_{33}$$

$$\Rightarrow A_{33} = 0 + 3 = 3.$$

Cofactor matrix is $\begin{bmatrix} 0 & -11 & 0 \\ 3 & 1 & -1 \\ 2 & 8 & 3 \end{bmatrix}.$

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\Rightarrow \text{adj}A = \begin{bmatrix} 0 & -11 & 0 \\ 3 & 1 & -1 \\ 2 & 8 & 3 \end{bmatrix}^T$$

$$\therefore \text{adj}A = \begin{bmatrix} 0 & 3 & 2 \\ -11 & 1 & 8 \\ 0 & -1 & 3 \end{bmatrix}$$

Now, multiplying A with its adjoint, we have:

$$\Rightarrow A(\text{adj}A) = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} 0 & 3 & 2 \\ -11 & 1 & 8 \\ 0 & -1 & 3 \end{bmatrix}$$

$$\Rightarrow A(\text{adj}A) = \begin{bmatrix} 0+11+0 & 3-1-2 & 2-8+6 \\ 0+0+0 & 9+0+2 & 6+0-6 \\ 0+0+0 & 3+0-3 & 2+0+9 \end{bmatrix}$$

$$\therefore A(\text{adj } A) = \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix}$$

Similarly, multiplying $(\text{adj } A)$ with A , we get:

$$\Rightarrow (\text{adj } A)A = \begin{bmatrix} 0 & 3 & 2 \\ -11 & 1 & 8 \\ 0 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{bmatrix}$$

$$\Rightarrow (\text{adj } A)A = \begin{bmatrix} 0+9+2 & 0+0+0 & 0-6+6 \\ -11+3+8 & 11+0+0 & -22-2+24 \\ 0-3+3 & 0+0+0 & 0+2+9 \end{bmatrix}$$

$$\therefore (\text{adj } A)A = \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix}$$

Thus, $A(\text{adj } A) = (\text{adj } A)A = |A|I$

Hence verified.

5. Find the inverse of each of the matrices (if it exists). $\begin{bmatrix} 2 & -2 \\ 4 & 3 \end{bmatrix}$

Ans: Let $A = \begin{bmatrix} 2 & -2 \\ 4 & 3 \end{bmatrix}$

$$\Rightarrow |A| = 6 + 8$$

$$\therefore |A| = 14$$

Since, Cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$.

Then,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = 3$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = -4$$

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = 2$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = 2$$

Cofactor matrix is $\begin{bmatrix} 3 & -4 \\ 2 & 2 \end{bmatrix}$.

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\therefore \text{adj}A = \begin{bmatrix} 3 & 2 \\ -4 & 2 \end{bmatrix}$$

Hence, the inverse of the matrix A is given by,

$$\Rightarrow A^{-1} = \frac{1}{|A|} \text{adj}A$$

$$\therefore A^{-1} = \frac{1}{14} \begin{bmatrix} 3 & 2 \\ -4 & 2 \end{bmatrix}.$$

6. Find the inverse of each of the matrices (if it exists). $\begin{bmatrix} -1 & 5 \\ -3 & 2 \end{bmatrix}$

Ans: Let $A = \begin{bmatrix} -1 & 5 \\ -3 & 2 \end{bmatrix}$

$$\Rightarrow |A| = -2 + 15$$

$$\therefore |A| = 13$$

Since, Cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$.

Then,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = 2$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = 3$$

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = -5$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = -1$$

Cofactor matrix is $\begin{bmatrix} 2 & 3 \\ -5 & -1 \end{bmatrix}$.

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\therefore \text{adj}A = \begin{bmatrix} 2 & -5 \\ 3 & -1 \end{bmatrix}$$

Hence, the inverse of the matrix A is given by,

$$\Rightarrow A^{-1} = \frac{1}{|A|} \text{adj}A$$

$$\therefore A^{-1} = \frac{1}{13} \begin{bmatrix} 2 & -5 \\ 3 & -1 \end{bmatrix}.$$

7. Find the inverse of each of the matrices (if it exists). $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 4 \\ 0 & 0 & 5 \end{bmatrix}$

Ans: Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 4 \\ 0 & 0 & 5 \end{bmatrix}$

Then,

$$\Rightarrow |A| = 1(10 - 0) - 2(0 - 0) + 3(0 - 0)$$

$$\therefore |A| = 10$$

Since, Cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$.

Thus,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = 10 - 0 = 10$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = -(0 + 0) = 0$$

$$\Rightarrow A_{13} = (-1)^{1+3} M_{13} = (-1)^4 M_{13}$$

$$\Rightarrow A_{13} = 0$$

Similarly,

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = -(10 - 0) = -10$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = 5 - 0 = 5$$

$$\Rightarrow A_{23} = (-1)^{2+3} M_{23} = (-1)^5 M_{23}$$

$$\Rightarrow A_{23} = -(0 - 0) = 0$$

And

$$\Rightarrow A_{31} = (-1)^{3+1} M_{31} = (-1)^4 M_{31}$$

$$\Rightarrow A_{31} = 8 - 6 = 2$$

$$\Rightarrow A_{32} = (-1)^{3+2} M_{32} = (-1)^5 M_{32}$$

$$\Rightarrow A_{32} = -(4 - 0) = -4$$

$$\Rightarrow A_{33} = (-1)^{3+3} M_{33} = (-1)^6 M_{33}$$

$$\Rightarrow A_{33} = 2 - 0 = 2.$$

Cofactor matrix is $\begin{bmatrix} 10 & 0 & 0 \\ -10 & 5 & 0 \\ 2 & -4 & 2 \end{bmatrix}.$

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\therefore \text{adj}A = \begin{bmatrix} 10 & -10 & 2 \\ 0 & 5 & -4 \\ 0 & 0 & 2 \end{bmatrix}$$

Hence, the inverse of the matrix A is given by,

$$\Rightarrow A^{-1} = \frac{1}{|A|} \text{adj}A$$

$$\therefore A^{-1} = \frac{1}{10} \begin{bmatrix} 10 & -10 & 2 \\ 0 & 5 & -4 \\ 0 & 0 & 2 \end{bmatrix}$$

8. Find the inverse of each of the matrices (if it exists). $\begin{bmatrix} 1 & 0 & 0 \\ 3 & 3 & 0 \\ 5 & 2 & -1 \end{bmatrix}$

Ans: Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 3 & 0 \\ 5 & 2 & -1 \end{bmatrix}$

Then,

$$\Rightarrow |A| = 1(-3-0)-0+0$$

$$\therefore |A| = -3$$

Since, Cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$.

Thus,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = -3 - 0 = -3$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = -(-3 - 0) = 3$$

$$\Rightarrow A_{13} = (-1)^{1+3} M_{13} = (-1)^4 M_{13}$$

$$\Rightarrow A_{13} = 6 - 15 = -9$$

Similarly,

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = -(0 + 0) = 0$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = -1 - 0 = -1$$

$$\Rightarrow A_{23} = (-1)^{2+3} M_{23} = (-1)^5 M_{23}$$

$$\Rightarrow A_{23} = -(2 - 0) = -2$$

and

$$\Rightarrow A_{31} = (-1)^{3+1} M_{31} = (-1)^4 M_{31}$$

$$\Rightarrow A_{31} = 0 - 0 = 0$$

$$\Rightarrow A_{32} = (-1)^{3+2} M_{32} = (-1)^5 M_{32}$$

$$\Rightarrow A_{32} = -(0 - 0) = 0$$

$$\Rightarrow A_{33} = (-1)^{3+3} M_{33} = (-1)^6 M_{33}$$

$$\Rightarrow A_{33} = 3 - 0 = 3.$$

Cofactor matrix is $\begin{bmatrix} -3 & 3 & -9 \\ 0 & -1 & -2 \\ 0 & 0 & 3 \end{bmatrix}$.

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\Rightarrow \text{adj}A = \begin{bmatrix} -3 & 3 & -9 \\ 0 & -1 & -2 \\ 0 & 0 & 3 \end{bmatrix}^T$$

$$\therefore \text{adj}A = \begin{bmatrix} -3 & 0 & 0 \\ 3 & -1 & 0 \\ -9 & -2 & 3 \end{bmatrix}$$

Hence, the inverse of the matrix A is given by,

$$\Rightarrow A^{-1} = \frac{1}{|A|} \text{adj}A$$

$$\therefore A^{-1} = \frac{1}{10} \begin{bmatrix} -3 & 0 & 0 \\ 3 & -1 & 0 \\ -9 & -2 & 3 \end{bmatrix}$$

9. Find the inverse of each of the matrices (if it exists). $\begin{bmatrix} 2 & 1 & 3 \\ 4 & -1 & 0 \\ -7 & 2 & 1 \end{bmatrix}$

Ans: Let $A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & -1 & 0 \\ -7 & 2 & 1 \end{bmatrix}$

Thus,

$$\Rightarrow |A| = 2(-1-0) - 1(4-0) + 3(8-7)$$

$$\Rightarrow |A| = 2(-1) - 1(4) + 3(1)$$

$$\therefore |A| = -3$$

Since, Cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$.

Thus,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = -1 - 0 = -1$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = -(4 - 0) = -4$$

$$\Rightarrow A_{13} = (-1)^{1+3} M_{13} = (-1)^4 M_{13}$$

$$\Rightarrow A_{13} = 8 - 7 = 1$$

Similarly,

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = -(1 - 6) = 5$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = 2 + 21 = 23$$

$$\Rightarrow A_{23} = (-1)^{2+3} M_{23} = (-1)^5 M_{23}$$

$$\Rightarrow A_{23} = -(4 + 7) = -11$$

and

$$\Rightarrow A_{31} = (-1)^{3+1} M_{31} = (-1)^4 M_{31}$$

$$\Rightarrow A_{31} = 0 + 3 = 3$$

$$\Rightarrow A_{32} = (-1)^{3+2} M_{32} = (-1)^5 M_{32}$$

$$\Rightarrow A_{32} = -(0 - 12) = 12$$

$$\Rightarrow A_{33} = (-1)^{3+3} M_{33} = (-1)^6 M_{33}$$

$$\Rightarrow A_{33} = -2 - 4 = -6.$$

Cofactor matrix is $\begin{bmatrix} -1 & -4 & 1 \\ 5 & 23 & -11 \\ 3 & 12 & -6 \end{bmatrix}.$

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\Rightarrow \text{adj}A = \begin{bmatrix} -1 & -4 & 1 \\ 5 & 23 & -11 \\ 3 & 12 & -6 \end{bmatrix}^T$$

$$\therefore \text{adj}A = \begin{bmatrix} -1 & 5 & 3 \\ -4 & 23 & 12 \\ 1 & -11 & -6 \end{bmatrix}$$

Hence, the inverse of the matrix A is given by,

$$\Rightarrow A^{-1} = \frac{1}{|A|} \text{adj}A$$

$$\therefore A^{-1} = \frac{1}{10} \begin{bmatrix} -1 & 5 & 3 \\ -4 & 23 & 12 \\ 1 & -11 & -6 \end{bmatrix}$$

10. Find the inverse of each of the matrices (if it exists). $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}$

Ans: Let $A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}$

Expanding along column C_1 ,

$$\Rightarrow |A| = 1(8 - 6) - 0 + 3(3 - 4)$$

$$\therefore |A| = -1$$

Since, Cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$.

Thus,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = 8 - 6 = 2$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = -(0 + 9) = -9$$

$$\Rightarrow A_{13} = (-1)^{1+3} M_{13} = (-1)^4 M_{13}$$

$$\Rightarrow A_{13} = 0 - 6 = -6$$

Similarly,

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = -(-4 + 4) = 0$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = 4 - 6 = -2$$

$$\Rightarrow A_{23} = (-1)^{2+3} M_{23} = (-1)^5 M_{23}$$

$$\Rightarrow A_{23} = -(-2 + 3) = -1$$

and

$$\Rightarrow A_{31} = (-1)^{3+1} M_{31} = (-1)^4 M_{31}$$

$$\Rightarrow A_{31} = 3 - 4 = -1$$

$$\Rightarrow A_{32} = (-1)^{3+2} M_{32} = (-1)^5 M_{32}$$

$$\Rightarrow A_{32} = -(-3 - 0) = 3$$

$$\Rightarrow A_{33} = (-1)^{3+3} M_{33} = (-1)^6 M_{33}$$

$$\Rightarrow A_{33} = 2 - 0 = 2.$$

Cofactor matrix is $\begin{bmatrix} 2 & -9 & -6 \\ 0 & -2 & -1 \\ -1 & 3 & 2 \end{bmatrix}.$

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\Rightarrow \text{adj}A = \begin{bmatrix} 2 & -9 & -6 \\ 0 & -2 & -1 \\ -1 & 3 & 2 \end{bmatrix}^T$$

$$\Rightarrow \text{adj}A = \begin{bmatrix} 2 & 0 & -1 \\ -9 & -2 & 3 \\ -6 & -1 & 2 \end{bmatrix}$$

The inverse of the matrix A is given by,

$$\Rightarrow A^{-1} = \frac{1}{|A|} \text{adj}A$$

$$\therefore A^{-1} = -1 \begin{bmatrix} 2 & 0 & -1 \\ -9 & -2 & 3 \\ -6 & -1 & 2 \end{bmatrix}$$

$$\text{Hence, } A^{-1} = \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}.$$

11. Find the inverse of each of the matrices (if it exists). $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos a & \sin a \\ 0 & \sin a & -\cos a \end{bmatrix}$

$$\text{Ans: Let } A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos a & \sin a \\ 0 & \sin a & -\cos a \end{bmatrix}$$

Expanding along column, C_1

$$\Rightarrow |A| = 1(-\cos^2 a - \sin^2 a)$$

$$\Rightarrow |A| = -(\cos^2 a + \sin^2 a)$$

$$\therefore |A| = -1$$

Since, Cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$.

Thus,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = -\cos^2 a - \sin^2 a = -1$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = 0$$

$$\Rightarrow A_{13} = (-1)^{1+3} M_{13} = (-1)^4 M_{13}$$

$$\Rightarrow A_{13} = 0$$

Similarly,

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = 0$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = -\cos a$$

$$\Rightarrow A_{23} = (-1)^{2+3} M_{23} = (-1)^5 M_{23}$$

$$\Rightarrow A_{23} = -\sin a$$

and

$$\Rightarrow A_{31} = (-1)^{3+1} M_{31} = (-1)^4 M_{31}$$

$$\Rightarrow A_{31} = 0$$

$$\Rightarrow A_{32} = (-1)^{3+2} M_{32} = (-1)^5 M_{32}$$

$$\Rightarrow A_{32} = -\sin a$$

$$\Rightarrow A_{33} = (-1)^{3+3} M_{33} = (-1)^6 M_{33}$$

$$\Rightarrow A_{33} = \cos a.$$

$$\text{Cofactor matrix is } \begin{bmatrix} -1 & 0 & 0 \\ 0 & -\cos a & -\sin a \\ 0 & -\sin a & \cos a \end{bmatrix}.$$

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\Rightarrow \text{adj}A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -\cos a & -\sin a \\ 0 & -\sin a & \cos a \end{bmatrix}^T$$

$$\therefore \text{adj}A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -\cos a & -\sin a \\ 0 & -\sin a & \cos a \end{bmatrix}$$

The inverse of the matrix A is given by,

$$\Rightarrow A^{-1} = \frac{1}{|A|} \text{adj}A$$

$$\therefore A^{-1} = -1 \begin{bmatrix} -1 & 0 & 0 \\ 0 & -\cos a & -\sin a \\ 0 & -\sin a & \cos a \end{bmatrix}$$

$$\text{Hence, } A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos a & \sin a \\ 0 & \sin a & -\cos a \end{bmatrix}.$$

12. Let $A = \begin{bmatrix} 3 & 7 \\ 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 6 & 8 \\ 7 & 9 \end{bmatrix}$. Verify that $(AB)^{-1} = B^{-1}A^{-1}$.

Ans: Let $A = \begin{bmatrix} 3 & 7 \\ 2 & 5 \end{bmatrix}$

Thus, determining the value of $|A|$,

$$\Rightarrow |A| = 15 - 14$$

$$\therefore |A| = 1$$

Since, Cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$.

Thus,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = 5$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = -2$$

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = -7$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = 3$$

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\therefore \text{adj}A = \begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix}^T$$

$$\Rightarrow \text{adj}A = \begin{bmatrix} 5 & -7 \\ -2 & 3 \end{bmatrix}$$

The inverse of a matrix is given by, $A^{-1} = \frac{1}{|A|} \text{adj}A$

$$\text{Hence, } A^{-1} = \begin{bmatrix} 5 & -7 \\ -2 & 3 \end{bmatrix}$$

$$\text{For } B = \begin{bmatrix} 6 & 8 \\ 7 & 9 \end{bmatrix}$$

$$\Rightarrow |B| = 54 - 56$$

$$\therefore |B| = -2$$

Thus,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = 9$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = -7$$

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = -8$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = 6$$

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\therefore \text{adj}A = \begin{bmatrix} 9 & -7 \\ -8 & 6 \end{bmatrix}^T$$

$$\Rightarrow \text{adj}A = \begin{bmatrix} 9 & -8 \\ -7 & 6 \end{bmatrix}$$

$$\text{Hence, } \text{adj}B = \begin{bmatrix} 9 & -8 \\ -7 & 6 \end{bmatrix}$$

$$\therefore B^{-1} = \frac{1}{|B|} \text{adj}B = -\frac{1}{2} \begin{bmatrix} 9 & -8 \\ -7 & 6 \end{bmatrix}$$

$$\text{Thus, } B^{-1} = \begin{bmatrix} -\frac{9}{2} & 4 \\ \frac{7}{2} & -3 \end{bmatrix}.$$

Now, multiplying B^{-1} and A^{-1} , we get:

$$\Rightarrow B^{-1}A^{-1} = \begin{bmatrix} -\frac{9}{2} & 4 \\ \frac{7}{2} & -3 \end{bmatrix} \begin{bmatrix} 5 & -7 \\ -2 & 3 \end{bmatrix}$$

$$\Rightarrow B^{-1}A^{-1} = \begin{bmatrix} -\frac{45}{2} - 8 & \frac{63}{2} + 12 \\ \frac{35}{2} + 6 & -\frac{49}{2} - 9 \end{bmatrix}$$

$$\therefore B^{-1}A^{-1} = \begin{bmatrix} -\frac{61}{2} & \frac{87}{2} \\ \frac{47}{2} & -\frac{67}{2} \end{bmatrix} \dots\dots(1)$$

Similarly, multiplying the matrices A and B, we get:

$$\Rightarrow AB = \begin{bmatrix} 3 & 7 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 6 & 8 \\ 7 & 9 \end{bmatrix}$$

$$\Rightarrow AB = \begin{bmatrix} 18+49 & 24+63 \\ 12+35 & 16+45 \end{bmatrix}$$

$$\therefore AB = \begin{bmatrix} 67 & 87 \\ 47 & 61 \end{bmatrix}$$

The value of $|AB|$ is

$$\Rightarrow |AB| = 67 \times 61 - 87 \times 47$$

$$\Rightarrow |AB| = 4087 - 4089$$

$$\therefore |AB| = -2$$

The adjoint of (AB) is given by,

$$\Rightarrow \text{adj}(AB) = \begin{bmatrix} 61 & -87 \\ -47 & 67 \end{bmatrix}$$

Thus, the inverse is,

$$\Rightarrow (AB)^{-1} = \frac{1}{|AB|} \text{adj}(AB)$$

$$\Rightarrow (AB)^{-1} = \frac{1}{-2} \begin{bmatrix} 61 & -87 \\ -47 & 67 \end{bmatrix}$$

$$\therefore (AB)^{-1} = \begin{bmatrix} -\frac{61}{2} & \frac{87}{2} \\ \frac{47}{2} & -\frac{67}{2} \end{bmatrix} \dots\dots (2)$$

From (1) and (2), we have:

$$(AB)^{-1} = B^{-1}A^{-1}$$

Hence proved.

13. If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, show that $A^2 - 5A + 7I = O$. Hence find A^{-1} .

Ans: Given, $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$

We can write, $A^2 = A.A$

$$\Rightarrow A^2 = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 9-1 & 3+2 \\ -3-2 & -1+4 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix}$$

$\therefore$ The value of $A^2 - 5A + 7I$ is:

$$\Rightarrow A^2 - 5A + 7I = \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix} - 5 \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} + 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow A^2 - 5A + 7I = \begin{bmatrix} -7 & 0 \\ 0 & -7 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$\therefore A^2 - 5A + 7I = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Hence, $A^2 - 5A + 7I = O$

$$\Rightarrow A^2 - 5A = -7I$$

Multiplying by A^{-1} on both the sides, we have:

$$\Rightarrow AA(A^{-1}) - 5AA^{-1} = -7IA^{-1}$$

$$\Rightarrow A(AA^{-1}) - 5I = -7IA^{-1}$$

$$\Rightarrow AI - 5I = -7IA^{-1}$$

$$\Rightarrow A^{-1} = -\frac{1}{7}(A - 5I)$$

$$\Rightarrow A^{-1} = \frac{1}{7}(5I - A)$$

$$\Rightarrow A^{-1} = \frac{1}{7} \left(\begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} - \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \right)$$

$$\therefore A^{-1} = \frac{1}{7} \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$$

14. For the matrix $A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$ find the number a and b such that.

$$A^2 + aA + bI = 0.$$

Ans: Given $A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$

We can write, $A^2 = A.A$

$$\therefore A^2 = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 9+2 & 6+2 \\ 3+1 & 2+1 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 11 & 8 \\ 4 & 3 \end{bmatrix}$$

Solving $A^2 + aA + bI = 0$ by multiplying the whole equation by A^{-1} .

$$\Rightarrow (AA)A^{-1} + aAA^{-1} + bIA^{-1} = 0$$

$$\Rightarrow A(AA^{-1}) + aI + b(IA^{-1}) = 0$$

$$\Rightarrow AI + aI + bA^{-1} = 0$$

$$\Rightarrow A + aI = -bA^{-1}$$

$$\Rightarrow A^{-1} = \frac{1}{b}(A + aI)$$

Now, determining the value of A^{-1} .

We know that the adjoint of a square matrix is the transpose of its cofactor matrix.

Hence, the adjoint of matrix A is:

$$\therefore \text{adj}A = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$$

The inverse is given by, $A^{-1} = \frac{1}{|A|} \text{adj}A$.

$$\Rightarrow A^{-1} = \frac{1}{1} \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$$

Thus,

$$\Rightarrow \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix} = -\frac{1}{b} \left(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} \right)$$

$$\Rightarrow \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix} = -\frac{1}{b} \begin{bmatrix} 3+a & 2 \\ 1 & 1+a \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} \frac{-3-a}{b} & -\frac{2}{b} \\ -\frac{1}{b} & \frac{-1-a}{b} \end{bmatrix}$$

Equating the corresponding elements of the two matrices, we get:

$$\Rightarrow -\frac{1}{b} = -1$$

$$\therefore b = 1$$

$$\Rightarrow \frac{-3-a}{b} = 1$$

$$\therefore a = -4$$

Thus, -4 and 1 are the required values of a and b respectively.

15. For the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$ show that $A^3 - 6A^2 + 5A + 11I = 0$. Hence, A^{-1} .

Ans: Given, $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$

$$\Rightarrow A^2 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 1+1+1 & 1+2-1 & 1-3+3 \\ 1+2-6 & 1+4+3 & 1-6-9 \\ 2-1+6 & 2-2-3 & 2+3+9 \end{bmatrix}$$

$$\therefore A^2 = \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix}$$

$$\Rightarrow A^3 = A^2 \cdot A = \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$$

$$\Rightarrow A^3 = \begin{bmatrix} 4+2+2 & 4+4-1 & 4-6+3 \\ -3+8-28 & -3+16+14 & -3-24-42 \\ 7-3+28 & 7-6-14 & 7+9+42 \end{bmatrix}$$

$$\therefore A^3 = \begin{bmatrix} 8 & 7 & 1 \\ -23 & 27 & -69 \\ 32 & -13 & 58 \end{bmatrix}$$

Substituting the values for A^3 , A^2 and A in $A^3 - 6A^2 + 5A + 11I$, we have:

$$\Rightarrow A^3 - 6A^2 + 5A + 11I =$$

$$\begin{bmatrix} 8 & 7 & 1 \\ -23 & 27 & -69 \\ 32 & -13 & 58 \end{bmatrix} - 6 \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix} + 5 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} + 11 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow A^3 - 6A^2 + 5A + 11I =$$

$$\begin{bmatrix} 8 & 7 & 1 \\ -23 & 27 & -69 \\ 32 & -13 & 58 \end{bmatrix} - \begin{bmatrix} 24 & 12 & 6 \\ -18 & 48 & -84 \\ 42 & -18 & 84 \end{bmatrix} + \begin{bmatrix} 5 & 5 & 5 \\ 5 & 10 & -15 \\ 2 & -5 & 15 \end{bmatrix} + 11 \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix}$$

$$\Rightarrow A^3 - 6A^2 + 5A + 11I = \begin{bmatrix} 24 & 12 & 6 \\ -18 & 48 & -84 \\ 42 & -18 & 84 \end{bmatrix} - \begin{bmatrix} 24 & 12 & 6 \\ -18 & 48 & -84 \\ 42 & -18 & 84 \end{bmatrix}$$

$$\therefore A^3 - 6A^2 + 5A + 11I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0$$

Thus, $A^3 - 6A^2 + 5A + 11I = 0$

Since, $A^3 - 6A^2 + 5A + 11I = 0$.

Multiplying the whole equation by A^{-1} , we have:

$$\Rightarrow (AAA)A^{-1} - 6(AA)A^{-1} + 5AA^{-1} + 11IA^{-1} = 0$$

$$\Rightarrow AA(AA^{-1}) - 6A(AA^{-1}) + 5(AA^{-1}) = 11(IA^{-1})$$

$$\Rightarrow A^2 - 6A + 5I = -11A^{-1}$$

$$\Rightarrow A^{-1} = -\frac{1}{11}(A^2 - 6A + 5I) \quad \dots (1)$$

Now, $A^2 - 6A + 5I$ is given by:

$$\Rightarrow A^2 - 6A + 5I = \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix} - 6 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} + 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow A^2 - 6A + 5I = \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix} - \begin{bmatrix} 6 & 6 & 6 \\ 6 & 12 & -18 \\ 12 & 6 & 18 \end{bmatrix} + \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$\Rightarrow A^2 - 6A + 5I = \begin{bmatrix} 4 & 2 & 1 \\ -3 & 13 & -14 \\ 7 & -3 & 19 \end{bmatrix} - \begin{bmatrix} 6 & 6 & 6 \\ 6 & 12 & -18 \\ 12 & -6 & 18 \end{bmatrix}$$

$$\therefore A^2 - 6A + 5I = \begin{bmatrix} 3 & -4 & -5 \\ -9 & 1 & 4 \\ -5 & 3 & 1 \end{bmatrix}$$

Substituting for $A^2 - 6A + 5I$ equation (1), we get

$$\Rightarrow A^{-1} = -\frac{1}{11} \begin{bmatrix} 3 & -4 & -5 \\ -9 & 1 & 4 \\ -5 & 3 & 1 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{11} \begin{bmatrix} -3 & 4 & 5 \\ 9 & -1 & -4 \\ 5 & -3 & -1 \end{bmatrix}$$

16. If $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ verify that $A^3 - 6A^2 + 9A + 4I = 0$ and hence find A^{-1} .

Ans: Given, $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$

$$\Rightarrow A^2 = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 4+1+1 & -2-2-1 & 2+1+2 \\ -2-2-1 & 1+4+1 & -1-2-2 \\ 2+1+2 & -1-2-2 & 1+1+4 \end{bmatrix}$$

$$\therefore A^2 = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}$$

Similarly,

$$\Rightarrow A^3 = A^2 A = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$\Rightarrow A^3 = \begin{bmatrix} 12+5+5 & -6-10-5 & 6+5+10 \\ -10-6-5 & 5+12+5 & -5-6-10 \\ 10+5+6 & -5-10-6 & 5+5+12 \end{bmatrix}$$

$$\therefore A^3 = \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix}$$

Now, $A^3 - 6A^2 + 9A - 4I$ is given by:

$$\Rightarrow A^3 - 6A^2 + 9A - 4I = \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix} - 6 \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} + 9 \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow A^3 - 6A^2 + 9A - 4I = \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix} - \begin{bmatrix} 36 & -30 & 30 \\ -30 & 36 & -30 \\ 30 & -30 & 36 \end{bmatrix} + \begin{bmatrix} 18 & -9 & 9 \\ -9 & 18 & -9 \\ 9 & -9 & 18 \end{bmatrix} - \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$\Rightarrow A^3 - 6A^2 + 9A - 4I = \begin{bmatrix} 40 & -30 & 30 \\ -30 & 40 & -30 \\ 30 & -30 & 40 \end{bmatrix} - \begin{bmatrix} 40 & -30 & 30 \\ -30 & 40 & -30 \\ 30 & -30 & 40 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore A^3 - 6A^2 + 9A - 4I = 0$$

Since, $A^3 - 6A^2 + 9A - 4I = 0$.

Multiplying the whole equation by A^{-1} , we have:

$$\Rightarrow (AAA)A^{-1} - 6(AA)A^{-1} + 9AA^{-1} - 4IA^{-1} = 0$$

$$\Rightarrow AA(AA^{-1}) - 6A(AA^{-1}) + 9(AA^{-1}) = 4(IA^{-1})$$

$$\Rightarrow AAI - 6AI + 9I = 4A^{-1}$$

$$\Rightarrow A^2 - 6A + 9I = 4A^{-1}$$

$$\Rightarrow A^{-1} = \frac{1}{4}(A^2 - 6A + 9I) \dots\dots (1)$$

Now, $A^2 - 6A + 9I$ is given by:

$$A^2 - 6A + 9I = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} - 6 \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} + 9 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A^2 - 6A + 9I = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} - \begin{bmatrix} 12 & -6 & 6 \\ -6 & 12 & -6 \\ 6 & -6 & 12 \end{bmatrix} + \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$\therefore A^2 - 6A + 9I = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 3 & 3 \end{bmatrix}$$

Substituting for $A^2 - 6A + 9I$ equation (1), we get

$$A^{-1} = \frac{1}{4} \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 3 & 3 \end{bmatrix}.$$

17. Let A be nonsingular square matrix of order 3×3 . Then $|\text{adj}A|$ is equal to

- A. $|A|$
- B. $|A|^2$
- C. $|A|^3$
- D. $3|A|$

Ans: Given A is a nonsingular square matrix, i.e., it is a square matrix whose determinant is not equal to zero.

The inverse of a matrix is given as $A^{-1} = \frac{1}{|A|} \text{adj}A$.

$$\Rightarrow A^{-1}A = \frac{1}{|A|} \text{adj}A$$

$$\Rightarrow |A|I = \text{adj}A$$

The adjoint of matrix A is given by,

$$\Rightarrow (\text{adj}A) = A = |A|I = \begin{bmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{bmatrix}$$

$$\Rightarrow |(\text{adj}A)A| = \begin{vmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{vmatrix}$$

$$\Rightarrow |\text{adj}A||A| = |A|^3 \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = |A|^3 (I)$$

$$\therefore |\text{adj}A| = |A|^2$$

Hence, **B.** $|A|^2$ is the correct answer.

18. If A is an invertible matrix of order 2, then $\det(A^{-1})$ is equal to

A. $\det(A)$

B. $\frac{1}{\det(A)}$

C. 1

D. 0

Ans: Since A is an invertible matrix, thus A^{-1} exists and it is given by:

$$A^{-1} = \frac{1}{|A|} \text{adj}A.$$

As matrix A is of order 2,

$$\therefore \text{Let } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Hence, $|A| = ad - bc$.

The adjoint of A would be, $\text{adj}A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

Now, the inverse of the matrix is given by:

$$\Rightarrow A^{-1} = \frac{1}{|A|} \text{adj}A$$

$$\Rightarrow A^{-1} = \begin{bmatrix} \frac{d}{|A|} & \frac{-b}{|A|} \\ \frac{-c}{|A|} & \frac{a}{|A|} \end{bmatrix}$$

$$\Rightarrow A^{-1} = \begin{bmatrix} \frac{d}{|A|} & \frac{-b}{|A|} \\ \frac{-c}{|A|} & \frac{a}{|A|} \end{bmatrix}$$

$$\Rightarrow |A^{-1}| = \frac{1}{|A^2|} \begin{vmatrix} d & -b \\ -c & a \end{vmatrix}$$

$$\Rightarrow |A^{-1}| = \frac{1}{|A^2|} (ad-bc)$$

$$\Rightarrow |A^{-1}| = \frac{1}{|A^2|} \cdot |A|$$

$$\therefore |A^{-1}| = \frac{1}{|A|}$$

$$\text{Thus, } \det(A^{-1}) = \frac{1}{\det(A)}.$$

Hence, **B.** $\frac{1}{\det(A)}$ is the correct answer.

Exercise 4.6

1. Examine the consistency of the system of equations.

$$x+2y=2$$

$$2x+3y=3$$

Ans: Given equations,

$$x+2y=2$$

$$2x+3y=3$$

Let us suppose $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$ and $B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ such that, the given system of equations can be written in the form of $AX=B$.

Determining the value of A , we have:

$$\Rightarrow |A| = 1(3) - 2(2) = 3 - 4$$

$$\therefore |A| = -1 \neq 0$$

Hence, A is non-singular.

Thus, the inverse of A , i.e., A^{-1} exists.

$\therefore$ The given system of equations is consistent.

2. Examine the consistency of the system of equations.

$$2x-y=5$$

$$x+y=4$$

Ans: Given equations,

$$2x-y=5$$

$$x+y=4$$

Let us suppose $A = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$ and $B = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$ such that, the given system of equation can be written in the form of $AX=B$.

Determining the value of A , we have:

$$\Rightarrow |A| = 2(1) - (-1)(1) = 2+1$$

$$\therefore |A| = 3 \neq 0$$

Hence, A is non-singular.

Thus, A^{-1} exists.

$\therefore$ The given system of equations is consistent.

3. Examine the consistency of the system of equations.

$$x+3y=5$$

$$2x+6y=8$$

Ans: Given equations,

$$x+3y=5$$

$$2x+6y=8$$

We know, that a given system of equations is consistent if it has at least one solution.

Let $A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$ and $B = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$ such that, the given system of equation can be written in the form of $AX=B$.

Determining the value of A , we have:

$$\Rightarrow |A| = 1(6) - 3(2)$$

$$\therefore |A| = 0$$

Hence, A is a singular matrix.

We know that the adjoint of a square matrix is the transpose of its cofactor matrix.

Determining the adjoint of the matrix A, we have:

$$\Rightarrow (\text{adj}A) = \begin{bmatrix} 6 & -3 \\ -2 & 1 \end{bmatrix}$$

$$\Rightarrow (\text{adj}A)B = \begin{bmatrix} 6 & -3 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 8 \end{bmatrix} = \begin{bmatrix} 30-24 \\ -10+8 \end{bmatrix}$$

$$\therefore (\text{adj}A)B = \begin{bmatrix} 6 \\ -2 \end{bmatrix} \neq 0$$

Thus, the solution of the given system of equations does not exist.

$\therefore$ The given system of equations is inconsistent.

4. Examine the consistency of the system of equations.

$$x+y+z=1$$

$$2x+3y+2z=2$$

$$ax+ay+2az=4$$

Ans: Given equations,

$$x+y+z=1$$

$$2x+3y+2z=2$$

$$ax+ay+2az=4$$

We know, that a given system of equations is consistent if it has at least one solution.

Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ a & a & 2a \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$ such that, the given system of equation

can be written in the form of $AX=B$.

Determining the value of A , we have:

$$\Rightarrow |A| = 1(6a - 2a) - 1(4a - 2a) + 1(2a - 3a)$$

$$\Rightarrow |A| = 4a - 2a - a$$

$$\therefore |A| = a \neq 0$$

Hence A is non-singular matrix.

Thus, A^{-1} exists.

$\therefore$ The given system of equation is consistent.

5. Examine the consistency of the system of equations.

$$3x - y - 2z = 2$$

$$2y - z = -1$$

$$3x - 5y = 3$$

Ans: Given equations,

$$3x - y - 2z = 2$$

$$2y - z = -1$$

$$3x - 5y = 3$$

We know, that a given system of equations is consistent if it has at least one solution.

Let $A = \begin{bmatrix} 3 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ such that, this system of equations can

be written in the form of $AX=B$.

Determining the value of A , we have:

$$\Rightarrow |A| = 3(-5) - 0 + 3(1 + 4)$$

$$\Rightarrow |A| = -15 + 15$$

$$\Rightarrow |A| = 0$$

$\therefore A$ is a singular matrix.

Determining the adjoint of matrix A .

Writing the cofactors,

$$A_{11} = (-1)^{1+1} \begin{vmatrix} 2 & -1 \\ -5 & 0 \end{vmatrix} = -5$$

$$A_{12} = (-1)^{1+2} \begin{vmatrix} 0 & -1 \\ 3 & 0 \end{vmatrix} = -3$$

$$A_{13} = (-1)^{1+3} \begin{vmatrix} 0 & 2 \\ 3 & -5 \end{vmatrix} = -6$$

$$A_{21} = (-1)^{2+1} \begin{vmatrix} -1 & -2 \\ -5 & 0 \end{vmatrix} = 10$$

$$A_{22} = (-1)^{2+2} \begin{vmatrix} 3 & -2 \\ 3 & 0 \end{vmatrix} = 6$$

$$A_{23} = (-1)^{2+3} \begin{vmatrix} 3 & -1 \\ 3 & -5 \end{vmatrix} = -[-15 + 3] = 12$$

$$A_{31} = (-1)^{3+1} \begin{vmatrix} -1 & -2 \\ 2 & -1 \end{vmatrix} = [1 + 4] = 5$$

$$A_{32} = (-1)^{3+2} \begin{vmatrix} 3 & -2 \\ 0 & -1 \end{vmatrix} = 3$$

$$A_{33} = (-1)^{3+3} \begin{vmatrix} 3 & -1 \\ 0 & 2 \end{vmatrix} = 6$$

Cofactor matrix is $\begin{bmatrix} -5 & -3 & -6 \\ 10 & 6 & 12 \\ 5 & 3 & 6 \end{bmatrix}$

Taking transpose,

$$\Rightarrow (\text{adj}A) = \begin{bmatrix} -5 & 10 & 5 \\ -3 & 6 & 3 \\ -6 & 12 & 6 \end{bmatrix}$$

$$\Rightarrow (\text{adj}A)B = \begin{bmatrix} -5 & 10 & 5 \\ -3 & 6 & 3 \\ -6 & 12 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} -10-10+15 \\ -6-6+9 \\ -12-12+18 \end{bmatrix}$$

$$\therefore (\text{adj}A)B = \begin{bmatrix} -5 \\ -3 \\ -6 \end{bmatrix} \neq 0$$

Thus, the solution of the given system of equation does not exist.

$\therefore$ The system of equations is inconsistent.

6. Examine the consistency of the system of equations.

$$5x - y + 4z = 5$$

$$2x + 3y + 5z = 2$$

$$5x - 2y + 6z = -1$$

Ans: Given equations,

$$5x - y + 4z = 5$$

$$2x + 3y + 5z = 2$$

$$5x - 2y + 6z = -1$$

Let $A = \begin{bmatrix} 5 & -1 & 4 \\ 2 & 3 & 5 \\ 3 & -2 & 6 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}$ such that, the system of equations can

be written in the form of $AX=B$.

Determining the value of A , we have:

$$\Rightarrow |A| = 5(18+10) + 1(12-25) + 4(-4-15)$$

$$\Rightarrow |A| = 5(28) + 1(-13) + 4(-19)$$

$$\Rightarrow |A| = 140 - 13 - 76$$

$$\therefore |A| = 51 \neq 0$$

Hence, A is a non-singular matrix.

Thus, A^{-1} exists.

$\therefore$ The given system of equations is consistent.

7. Solve system of linear equations, using matrix method.

$$5x+2y=4$$

$$7x+3y=5$$

Ans: Given equations,

$$5x+2y=4$$

$$7x+3y=5$$

Let $A = \begin{bmatrix} 5 & 2 \\ 7 & 3 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$ and $B = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$ such that, this system of equations can be written in the form of $AX=B$.

Determining the value of A , we have:

$$\Rightarrow |A| = 15 - 14$$

$$\therefore |A| = 1 \neq 0$$

Thus, A is a non-singular matrix.

Hence, its inverse exists.

We know that the adjoint of a square matrix is the transpose of its cofactor matrix.

$$\therefore \text{adj}A = \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix}$$

Thus,

$$\Rightarrow A^{-1} = \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix}$$

$$\Rightarrow X = A^{-1}B = \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 12+10 \\ -28+52 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

Thus, $x=2$ and $y=-3$.

8. Solve system of linear equations, using matrix method.

$$2x-y=-2$$

$$3x+4y=3$$

Ans: Given equations,

$$2x-y=-2$$

$$3x+4y=3$$

Let $A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$ and $B = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$ such that, this system of equations can be written in the form of $AX=B$.

Determining the value of A , we have:

$$\Rightarrow |A| = 8+3$$

$$\therefore |A| = 11 \neq 0$$

Thus, A is non-singular.

$\therefore$ It's inverse exists.

We know that the adjoint of a square matrix is the transpose of its cofactor matrix.

$$\therefore \text{adj}A = \begin{bmatrix} 4 & 1 \\ -3 & 2 \end{bmatrix}$$

Now,

$$\Rightarrow A^{-1} = \frac{1}{|A|}(\text{adj}A)$$

$$\Rightarrow A^{-1} = \frac{1}{11} \begin{bmatrix} 4 & 1 \\ -3 & 2 \end{bmatrix}$$

$$\therefore X = A^{-1}B = \frac{1}{11} \begin{bmatrix} 4 & 1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{11} \begin{bmatrix} -8+3 \\ 6+6 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{11} \begin{bmatrix} -5 \\ 12 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -\frac{5}{11} \\ \frac{12}{11} \end{bmatrix}$$

Thus, $x = -\frac{5}{11}$ and $y = \frac{12}{11}$.

9. Solve system of linear equations, using matrix method.

$$4x-3y=3$$

$$3x-5y=7$$

Ans: Given equations,

$$4x-3y=3$$

$$3x-5y=7.$$

Let $A = \begin{bmatrix} 4 & -3 \\ 3 & -5 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$ and $B = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$ such that, this system of equations can be written in the form of $AX=B$.

Determining the value of A , we have:

$$\Rightarrow |A| = -20 + 9$$

$$\therefore |A| = -11 \neq 0$$

Thus, A is a non-singular matrix.

Hence, its inverse exists.

Formula for inverse is $A^{-1} = \frac{\text{adj}A}{|A|}$.

Finding cofactors,

$$A_{11} = (-1)^{1+1}(-5) = -5$$

$$A_{12} = (-1)^{1+2}(3) = -3$$

$$A_{21} = (-1)^{2+1}(-3) = 3$$

$$A_{22} = (-1)^{2+2}(4) = 4$$

Cofactor matrix is $\begin{bmatrix} -5 & -3 \\ 3 & 4 \end{bmatrix}$.

Taking its transpose to get adjoint matrix as $\begin{bmatrix} -5 & 3 \\ -3 & 4 \end{bmatrix}$.

Therefore inverse is

$$\Rightarrow A^{-1} = -\frac{1}{11} \begin{bmatrix} -5 & 3 \\ -3 & 4 \end{bmatrix}$$

$$\therefore X = A^{-1}B = -\frac{1}{11} \begin{bmatrix} -5 & 3 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ 7 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 5 & -3 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} 3 \\ 7 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 15-21 \\ 9-28 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -\frac{6}{11} \\ -\frac{19}{11} \end{bmatrix}$$

Thus, $x = -\frac{6}{11}$ and $y = -\frac{19}{11}$.

10. Solve system of linear equations, using matrix method.

$$5x+2y=3$$

$$3x+2y=5$$

Ans: Given equations,

$$5x+2y=3$$

$$3x+2y=5$$

Let $A = \begin{bmatrix} 5 & 2 \\ 3 & 2 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$ and $B = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$ such that, this system of equations can be written in the form of $AX=B$.

Determining the value of A , we have:

$$\Rightarrow |A| = 10 - 6$$

$$\therefore |A| = 4 \neq 0$$

Thus A is non-singular,

Therefore, its inverse exists.

Formula for inverse is $A^{-1} = \frac{\text{adj}A}{|A|}$.

Finding cofactors,

$$A_{11} = (-1)^{1+1}(2) = 2$$

$$A_{12} = (-1)^{1+2}(3) = -3$$

$$A_{21} = (-1)^{2+1}(2) = -2$$

$$A_{22} = (-1)^{2+2}(5) = 5$$

Cofactor matrix is $\begin{bmatrix} 2 & -3 \\ -2 & 5 \end{bmatrix}$.

Taking its transpose to get adjoint matrix as $\begin{bmatrix} 2 & -2 \\ -3 & 5 \end{bmatrix}$.

Therefore inverse is

$$\Rightarrow A^{-1} = \frac{1}{4} \begin{bmatrix} 2 & -2 \\ -3 & 5 \end{bmatrix}$$

$$\therefore X = A^{-1}B = \frac{1}{4} \begin{bmatrix} 2 & -2 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 2 & -2 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 6-10 \\ -9+25 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -\frac{4}{4} \\ \frac{16}{4} \end{bmatrix}$$

Thus, $x = -1$ and $y = 4$.

11. Solve system of linear equations, using matrix method.

$$2x+y+z=1$$

$$x-2y-z=\frac{3}{2}$$

$$3y-5z=9$$

Ans: Given equations,

$$2x+y+z=1$$

$$x-2y-z=\frac{3}{2}$$

$$3y-5z=9$$

Let $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & -2 & -1 \\ 0 & 3 & -5 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ \frac{3}{2} \\ 9 \end{bmatrix}$ such that, this system of equations can

be written in the form of $AX=B$.

The determinant of A is found by expanding along the first column,

$$\Rightarrow |A| = 2(10+3) - 1(-5-3) + 0$$

$$\Rightarrow |A| = 2(13) - 1(-8)$$

$$\Rightarrow |A| = 34 \neq 0$$

Thus, A is a non-singular matrix.

$\therefore$ Its inverse exists.

Hence,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = 13$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = 5$$

$$\Rightarrow A_{13} = (-1)^{1+3} M_{13} = (-1)^4 M_{13}$$

$$\Rightarrow A_{13} = 3$$

Similarly,

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = 8$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = -10$$

$$\Rightarrow A_{23} = (-1)^{2+3} M_{23} = (-1)^5 M_{23}$$

$$\Rightarrow A_{23} = -6$$

and

$$\Rightarrow A_{31} = (-1)^{3+1} M_{31} = (-1)^4 M_{31}$$

$$\Rightarrow A_{31} = 1$$

$$\Rightarrow A_{32} = (-1)^{3+2} M_{32} = (-1)^5 M_{32}$$

$$\Rightarrow A_{32} = 3$$

$$\Rightarrow A_{33} = (-1)^{3+3} M_{33} = (-1)^6 M_{33}$$

$$\Rightarrow A_{33} = -5.$$

Cofactor matrix is $\begin{bmatrix} 13 & 5 & 3 \\ 8 & -10 & -6 \\ 1 & 3 & -5 \end{bmatrix}.$

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\Rightarrow \text{adj}A = \begin{bmatrix} 13 & 5 & 3 \\ 8 & -10 & -6 \\ 1 & 3 & -5 \end{bmatrix}^T$$

$$\Rightarrow \text{adj}A = \begin{bmatrix} 13 & 8 & 1 \\ 5 & -10 & 3 \\ 3 & -16 & -5 \end{bmatrix}$$

The inverse of a matrix is given by:

$$\Rightarrow A^{-1} = \frac{1}{|A|} (\text{adj}A)$$

$$\therefore A^{-1} = \frac{1}{34} \begin{bmatrix} 13 & 8 & 1 \\ 5 & -10 & 3 \\ 3 & -16 & -5 \end{bmatrix}$$

$$\therefore X = A^{-1}B = \frac{1}{34} \begin{bmatrix} 13 & 8 & 1 \\ 5 & -10 & 3 \\ 3 & -16 & -5 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 2 \\ 9 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{34} \begin{bmatrix} 13+12+9 \\ 5-15+27 \\ 3-9-45 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{34} \begin{bmatrix} 34 \\ 17 \\ -51 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{1}{2} \\ -\frac{3}{2} \end{bmatrix}$$

Thus, $x = 1$ and $y = \frac{1}{2}$ and $z = -\frac{3}{2}$.

12. Solve system of linear equations, using matrix method.

$$x - y + z = 4$$

$$2x + y - 3z = 0$$

$$x + y + z = 2$$

Ans: Given equations,

$$x - y + z = 4$$

$$2x+y-3z=0$$

$$x+y+z=2$$

Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$ such that, this system of equations can be

written in the form of $AX=B$.

Determining the value of A , we have:

$$\Rightarrow |A| = 1(1+3) + 1(2+3) + 1(2-1)$$

$$\Rightarrow |A| = 4+5+1$$

$$\therefore |A| = 10 \neq 0$$

Thus A is non-singular.

$\therefore$ Its inverse exists.

Hence,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = 4$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = -5$$

$$\Rightarrow A_{13} = (-1)^{1+3} M_{13} = (-1)^4 M_{13}$$

$$\Rightarrow A_{13} = 1$$

Similarly,

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = 2$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = 0$$

$$\Rightarrow A_{23} = (-1)^{2+3} M_{23} = (-1)^5 M_{23}$$

$$\Rightarrow A_{23} = -2$$

and

$$\Rightarrow A_{31} = (-1)^{3+1} M_{31} = (-1)^4 M_{31}$$

$$\Rightarrow A_{31} = 2$$

$$\Rightarrow A_{32} = (-1)^{3+2} M_{32} = (-1)^5 M_{32}$$

$$\Rightarrow A_{32} = 5$$

$$\Rightarrow A_{33} = (-1)^{3+3} M_{33} = (-1)^6 M_{33}$$

$$\Rightarrow A_{33} = 3.$$

Cofactor matrix is $\begin{bmatrix} 4 & -5 & 1 \\ 2 & 0 & -2 \\ 2 & 5 & 3 \end{bmatrix}.$

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\Rightarrow \text{adj}A = \begin{bmatrix} 4 & -5 & 1 \\ 2 & 0 & -2 \\ 2 & 5 & 3 \end{bmatrix}^T$$

The inverse of a matrix is given by:

$$\Rightarrow A^{-1} = \frac{1}{|A|}(\text{adj}A)$$

$$\therefore A^{-1} = \frac{1}{10} \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$$

$$\therefore X = A^{-1}B$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 16+0+4 \\ -20+0+10 \\ 4+0+6 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 20 \\ -10 \\ 10 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

Thus, $x = 2$, $y = -1$ and $z = 1$.

13. Solve system of linear equations, using matrix method.

$$2x+3y+3z=5$$

$$x-2y+z=-4$$

$$3x-y-2z=3$$

Ans: Given equations,

$$2x+3y+3z=5$$

$$x-2y+z=-4$$

$$3x-y-2z=3$$

Let $A = \begin{bmatrix} 2 & 3 & 3 \\ 1 & -2 & 1 \\ 3 & -1 & -2 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix}$ such that, this system of equations can

be written in the form of $AX=B$.

Determining the value of A , we have:

$$\Rightarrow |A| = 2(4+1) - 3(2-3) + 3(-1+6)$$

$$\Rightarrow |A| = 2(5) - 3(-5) + 3(5)$$

$$\Rightarrow |A| = 10 + 15 + 15$$

$$\Rightarrow |A| = 40 \neq 0$$

Thus, A is a non-singular matrix.

$\therefore$ Its inverse exists.

Hence,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = 5$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = 5$$

$$\Rightarrow A_{13} = (-1)^{1+3} M_{13} = (-1)^4 M_{13}$$

$$\Rightarrow A_{13} = 5$$

Similarly,

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = 3$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = -13$$

$$\Rightarrow A_{23} = (-1)^{2+3} M_{23} = (-1)^5 M_{23}$$

$$\Rightarrow A_{23} = 11$$

and

$$\Rightarrow A_{31} = (-1)^{3+1} M_{31} = (-1)^4 M_{31}$$

$$\Rightarrow A_{31} = 9$$

$$\Rightarrow A_{32} = (-1)^{3+2} M_{32} = (-1)^5 M_{32}$$

$$\Rightarrow A_{32} = 1$$

$$\Rightarrow A_{33} = (-1)^{3+3} M_{33} = (-1)^6 M_{33}$$

$$\Rightarrow A_{33} = -7.$$

Cofactor matrix is $\begin{bmatrix} 5 & 5 & 5 \\ 3 & -13 & 11 \\ 9 & 1 & -7 \end{bmatrix}.$

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\Rightarrow \text{adj}A = \begin{bmatrix} 5 & 5 & 5 \\ 3 & -13 & 11 \\ 9 & 1 & -7 \end{bmatrix}^T$$

The inverse of a matrix is given by:

$$\Rightarrow A^{-1} = \frac{1}{|A|}(\text{adj}A)$$

$$\therefore A^{-1} = \frac{1}{40} \begin{bmatrix} 5 & 3 & 9 \\ 5 & -13 & 1 \\ 5 & 11 & -7 \end{bmatrix}$$

$$\therefore X = A^{-1}B$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{40} \begin{bmatrix} 5 & 3 & 9 \\ 5 & -13 & 1 \\ 5 & 11 & -7 \end{bmatrix} \begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{40} \begin{bmatrix} 25-12+27 \\ 25+52+3 \\ 25-44-21 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{40} \begin{bmatrix} 40 \\ 80 \\ -40 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

Thus, $x = 1$, $y = 2$ and $z = -1$.

14. Solve system of linear equations, using matrix method.

$$x-y+2z=7$$

$$3x+4y-5z=-5$$

$$2x-y+3z=12$$

Ans: Given equations,

$$x-y+2z=7$$

$$3x+4y-5z=-5$$

$$2x-y+3z=12$$

Let $A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 4 & -5 \\ 2 & -1 & 3 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 7 \\ -5 \\ 12 \end{bmatrix}$ such that, this system of equations can

be written in the form of $AX=B$.

Determining the value of A , we have:

$$\Rightarrow |A| = 1(12-5) + 1(9+10) + 2(-3-8)$$

$$\Rightarrow |A| = 7 + 19 - 22$$

$$\therefore |A| = 4 \neq 0$$

Thus, A is a non-singular matrix.

$\therefore$ Its inverse exists.

Now,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = 7$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = -19$$

$$\Rightarrow A_{13} = (-1)^{1+3} M_{13} = (-1)^4 M_{13}$$

$$\Rightarrow A_{13} = -11$$

Similarly,

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = 1$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = -1$$

$$\Rightarrow A_{23} = (-1)^{2+3} M_{23} = (-1)^5 M_{23}$$

$$\Rightarrow A_{23} = -1$$

and

$$\Rightarrow A_{31} = (-1)^{3+1} M_{31} = (-1)^4 M_{31}$$

$$\Rightarrow A_{31} = -3$$

$$\Rightarrow A_{32} = (-1)^{3+2} M_{32} = (-1)^5 M_{32}$$

$$\Rightarrow A_{32} = 11$$

$$\Rightarrow A_{33} = (-1)^{3+3} M_{33} = (-1)^6 M_{33}$$

$$\Rightarrow A_{33} = 7.$$

Cofactor matrix is $\begin{bmatrix} 7 & -19 & -11 \\ 1 & -1 & -1 \\ -3 & 11 & 7 \end{bmatrix}.$

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\Rightarrow \text{adj}A = \begin{bmatrix} 7 & -19 & -11 \\ 1 & -1 & -1 \\ -3 & 11 & 7 \end{bmatrix}^T$$

$$\Rightarrow \text{adj}A = \begin{bmatrix} 7 & 1 & -3 \\ -19 & -1 & 11 \\ -11 & -1 & 7 \end{bmatrix}$$

The inverse of a matrix is given by:

$$\Rightarrow A^{-1} = \frac{1}{|A|}(\text{adj}A)$$

$$\Rightarrow A^{-1} = \frac{1}{4} \begin{bmatrix} 7 & 1 & -3 \\ -19 & -1 & 11 \\ -11 & -1 & 7 \end{bmatrix}$$

$$\therefore X = A^{-1}B$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 7 & 1 & -3 \\ -19 & -1 & 11 \\ -11 & -1 & 7 \end{bmatrix} \begin{bmatrix} 7 \\ -5 \\ 12 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 49-5-36 \\ -133+5+132 \\ -77+5+84 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 8 \\ 4 \\ 12 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

Thus $x=2$, $y=1$ and $z=3$

15. If $A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}$, find A^{-1} . Using A^{-1} solve the system of equations

$$2x - 3y + 5z = 11$$

$$3x + 2y - 4z = -5$$

$$x + y - 2z = -3$$

Ans: Given equations,

$$2x - 3y + 5z = 11$$

$$3x + 2y - 4z = -5$$

$$x + y - 2z = -3$$

Let $A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 11 \\ -5 \\ -3 \end{bmatrix}$ such that, this system of equations

can be written in the form of $AX=B$.

Determining the value of A , we have:

$$\Rightarrow |A| = 2(-4+4) + 3(-6+4) + 5(3-2)$$

$$\Rightarrow |A| = 0 - 6 + 5$$

$$\therefore |A| = -1 \neq 0$$

Thus, A is a non-singular matrix.

∴ It's inverse exists.

Now,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = 0$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = 2$$

$$\Rightarrow A_{13} = (-1)^{1+3} M_{13} = (-1)^4 M_{13}$$

$$\Rightarrow A_{13} = 1$$

Similarly,

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = -1$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = -9$$

$$\Rightarrow A_{23} = (-1)^{2+3} M_{23} = (-1)^5 M_{23}$$

$$\Rightarrow A_{23} = -5$$

and

$$\Rightarrow A_{31} = (-1)^{3+1} M_{31} = (-1)^4 M_{31}$$

$$\Rightarrow A_{31} = 2$$

$$\Rightarrow A_{32} = (-1)^{3+2} M_{32} = (-1)^5 M_{32}$$

$$\Rightarrow A_{32} = 23$$

$$\Rightarrow A_{33} = (-1)^{3+3} M_{33} = (-1)^6 M_{33}$$

$$\Rightarrow A_{33} = 13.$$

Cofactor matrix is $\begin{bmatrix} 0 & 2 & 1 \\ -1 & -9 & -5 \\ 2 & 23 & 13 \end{bmatrix}.$

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\Rightarrow \text{adj}A = \begin{bmatrix} 0 & 2 & 1 \\ -1 & -9 & -5 \\ 2 & 23 & 13 \end{bmatrix}^T$$

$$\Rightarrow \text{adj}A = \begin{bmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{bmatrix}$$

The inverse of a matrix is given by:

$$\Rightarrow A^{-1} = \frac{1}{|A|} (\text{adj}A)$$

$$\Rightarrow A^{-1} = \frac{1}{-1} \begin{bmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{bmatrix}$$

$$\therefore X = A^{-1}B$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = -1 \begin{bmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{bmatrix} \begin{bmatrix} 11 \\ -5 \\ -3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = -1 \begin{bmatrix} 5-6 \\ 22+45-69 \\ 11+25-39 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = -1 \begin{bmatrix} -1 \\ -2 \\ -3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Thus $x=1$, $y=2$ and $z=3$.

16. The cost of 4kg onion, 3kg wheat and 2kg rice is Rs60. The cost of 2kg onion, 4kg wheat and 6kg rice is Rs90. The cost of 6kg onion 2kg wheat and 3kg rice is Rs70.

Find cost of each item per kg by matrix method.

Ans: Let us suppose that the cost of onions, wheat and rice per kg be Rs x , Rs y and Rs z respectively.

Then, the given situation can be represented by a system of equations as:

$$\Rightarrow 4x+3y+2z=60$$

$$\Rightarrow 2x+4y+6z=90$$

$$\Rightarrow 6x+2y+3z=70$$

Let $A = \begin{bmatrix} 4 & 3 & 2 \\ 2 & 4 & 6 \\ 6 & 2 & 3 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 60 \\ 90 \\ 70 \end{bmatrix}$ such that, this system of equations can

be written in the form of $AX=B$.

$$\Rightarrow |A| = 4(12-12) - 3(6-36) + 2(4-24)$$

$$\Rightarrow |A| = 0 + 90 - 40$$

$$\Rightarrow |A| = 50 \neq 0$$

Thus,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = 0$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = 30$$

$$\Rightarrow A_{13} = (-1)^{1+3} M_{13} = (-1)^4 M_{13}$$

$$\Rightarrow A_{13} = -20$$

Similarly,

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = -5$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = 0$$

$$\Rightarrow A_{23} = (-1)^{2+3} M_{23} = (-1)^5 M_{23}$$

$$\Rightarrow A_{23} = 10$$

And

$$\Rightarrow A_{31} = (-1)^{3+1} M_{31} = (-1)^4 M_{31}$$

$$\Rightarrow A_{31} = 10$$

$$\Rightarrow A_{32} = (-1)^{3+2} M_{32} = (-1)^5 M_{32}$$

$$\Rightarrow A_{32} = -20$$

$$\Rightarrow A_{33} = (-1)^{3+3} M_{33} = (-1)^6 M_{33}$$

$$\Rightarrow A_{33} = 10.$$

Cofactor matrix is $\begin{bmatrix} 0 & 30 & -20 \\ -5 & 0 & 10 \\ 10 & -20 & 10 \end{bmatrix}.$

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\Rightarrow \text{adj}A = \begin{bmatrix} 0 & 30 & -20 \\ -5 & 0 & 10 \\ 10 & -20 & 10 \end{bmatrix}^T$$

$$\therefore (\text{adj}A) = \begin{bmatrix} 0 & -5 & 10 \\ 30 & 0 & -20 \\ -20 & 10 & 10 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{1}{|A|} (\text{adj}A)$$

$$\therefore A^{-1} = \frac{1}{50} \begin{bmatrix} 0 & -5 & 10 \\ 30 & 0 & -20 \\ -20 & 10 & 10 \end{bmatrix}$$

Since, $X = A^{-1}B$

$$\Rightarrow X = \frac{1}{50} \begin{bmatrix} 0 & -5 & 10 \\ 30 & 0 & -20 \\ -20 & 10 & 10 \end{bmatrix} \begin{bmatrix} 60 \\ 90 \\ 70 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{50} \begin{bmatrix} 0+450+700 \\ 1800+0-1400 \\ -1200+900+700 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{50} \begin{bmatrix} 250 \\ 400 \\ 400 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \\ 8 \end{bmatrix}$$

Thus $x=5$, $y=8$, and $z=8$

Hence, the cost of onions is Rs 5 per kg, the cost of wheat is Rs 8 per kg, and the cost of rice is Rs 8 per kg.

Miscellaneous Solutions

1. Prove that the determinant $\begin{vmatrix} x & \sin \theta & \cos \theta \\ -\sin \theta & -x & 1 \\ \cos \theta & 1 & x \end{vmatrix}$ is independent of θ .

Ans: Let $\Delta = \begin{vmatrix} x & \sin \theta & \cos \theta \\ -\sin \theta & -x & 1 \\ \cos \theta & 1 & x \end{vmatrix}$

Solving it, we have:

$$\Rightarrow \Delta = x(x^2 - 1) - \sin \theta(-x \sin \theta - \cos \theta) + \cos \theta(-\sin \theta + x \cos \theta)$$

$$\Rightarrow \Delta = x^3 - x + x \sin^2 \theta + \sin \theta \cos \theta - \sin \theta \cos \theta + x \cos^2 \theta$$

$$\Rightarrow \Delta = x^3 - x + x(\sin^2 \theta + \cos^2 \theta)$$

$$\Rightarrow \Delta = x^3 - x + x$$

$$\therefore \Delta = x^3$$

Hence, Δ is independent of θ .

2. Without expanding the determinant, prove that

$$\begin{vmatrix} a & a^2 & bc \\ b & b^2 & ca \\ c & c^2 & ab \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}.$$

Ans: Solving the left-hand side determinant.

$$\text{L.H.S.} = \begin{vmatrix} a & a^2 & bc \\ b & b^2 & ca \\ c & c^2 & ab \end{vmatrix}$$

Applying the row operations, $R_1 \rightarrow aR_1$, $R_2 \rightarrow bR_2$ and $R_3 \rightarrow cR_3$

$$\Rightarrow \frac{1}{abc} \begin{vmatrix} a^2 & a^3 & abc \\ b^2 & b^3 & abc \\ c^2 & c^3 & abc \end{vmatrix}$$

Taking out abc common from C_3

$$\Rightarrow \frac{1}{abc} abc \begin{vmatrix} a^2 & a^3 & 1 \\ b^2 & b^3 & 1 \\ c^2 & c^3 & 1 \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} a^2 & a^3 & 1 \\ b^2 & b^3 & 1 \\ c^2 & c^3 & 1 \end{vmatrix}$$

Performing the interchange of columns, $C_1 \leftrightarrow C_3$ and $C_2 \leftrightarrow C_3$

$$\Rightarrow \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = \text{R.H.S}$$

Thus, L.H.S = R.H.S.

$$\begin{vmatrix} a & a^2 & bc \\ b & b^2 & ca \\ c & c^2 & ab \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$$

Hence proved.

3. Evaluate $\begin{vmatrix} \cos \alpha \cos \beta & \cos \alpha \sin \beta & -\sin \alpha \\ -\sin \beta & \cos \beta & 0 \\ \sin \alpha \cos \beta & \sin \alpha \sin \beta & \cos \alpha \end{vmatrix}$

Ans: Let $\Delta = \begin{vmatrix} \cos \alpha \cos \beta & \cos \alpha \sin \beta & -\sin \alpha \\ -\sin \beta & \cos \beta & 0 \\ \sin \alpha \cos \beta & \sin \alpha \sin \beta & \cos \alpha \end{vmatrix}$

Expanding along column C_3

$$\Rightarrow \Delta = -\sin \alpha (-\sin \alpha \sin^2 \beta + \cos^2 \beta \sin \alpha) + \cos \alpha (\cos \alpha \cos^2 \beta + \cos \alpha \sin^2 \beta)$$

$$\Rightarrow \Delta = \sin^2 \alpha (\sin^2 \beta + \cos^2 \beta) + \cos^2 \alpha (\cos^2 \beta + \sin^2 \beta)$$

$$\Rightarrow \Delta = \sin^2 \alpha (1) + \cos^2 \alpha (1)$$

$$\therefore \Delta = 1$$

4. If a, b and c are real numbers, and $\Delta = \begin{vmatrix} b+c & c+a & a+b \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix} = 0$

Show that either $a+b+c=0$ or $a=b=c$.

Ans: Given, $\Delta = \begin{vmatrix} b+c & c+a & a+b \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix} = 0$

Applying the row operation, $R_1 \rightarrow R_1 + R_2 + R_3$

$$\Rightarrow \Delta = \begin{vmatrix} 2(a+b+c) & 2(a+b+c) & 2(a+b+c) \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix}$$

Taking out $2(a+b+c)$ common from R_1 .

$$\Rightarrow \Delta = 2(a+b+c) \begin{vmatrix} 1 & 1 & 1 \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix}$$

Applying the row operations $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$

$$\Rightarrow \Delta = 2(a+b+c) \begin{vmatrix} 1 & 0 & 0 \\ c+a & b-c & b-a \\ a+b & c-a & c-b \end{vmatrix}$$

Expanding along R_1

$$\Rightarrow \Delta = 2(a+b+c)(1)[(b-c)(c-b) - (b-a)(c-a)]$$

$$\Rightarrow \Delta = 2(a+b+c)[-b^2 - c^2 + 2bc - bc + ba + ac - a^2]$$

$$\Rightarrow \Delta = 2(a+b+c)[ab + bc + ca - a^2 - b^2 - c^2]$$

It is given that $\Delta=0$.

$$\text{Thus, } (a+b+c)[ab+bc+ca-a^2-b^2-c^2]=0$$

$\therefore$ Either $a+b+c=0$, or $ab+bc+ca-a^2-b^2-c^2=0$.

$$\text{Now, } ab+bc+ca-a^2-b^2-c^2=0$$

$$\Rightarrow -2ab-2bc-2ca+2a^3+2b^3+2c^3=0$$

$$\Rightarrow (a-b)^2 + (b-c)^2 + (c-a)^2 = 0$$

$$\Rightarrow (a-b) = (b-c) = (c-a) = 0$$

Since, $(a-b)^2, (b-c)^2, (c-a)^2$ have non-negative values.

$$\therefore (a-b) = (b-c) = (c-a) = 0$$

$$\Rightarrow a=b=c$$

Thus, if $\Delta=0$, then either $a+b+c=0$ or $a=b=c$.

5. Solve the equations $\begin{vmatrix} x+a & x & x \\ x & x+a & x \\ x & x & x+a \end{vmatrix} = 0$, $a \neq 0$.

Ans: Given, $\begin{vmatrix} x+a & x & x \\ x & x+a & x \\ x & x & x+a \end{vmatrix} = 0$

Applying $R_1 \rightarrow R_1 + R_2 + R_3$

$$\Rightarrow \begin{vmatrix} 3x+a & 3x+a & 3x+a \\ x & x+a & x \\ x & x & x+a \end{vmatrix} = 0$$

Taking out $(3x + a)$ common from R_1 ,

$$\Rightarrow (3x+a) \begin{vmatrix} 1 & 1 & 1 \\ x & x+a & x \\ x & x & x+a \end{vmatrix} = 0$$

Applying the row operations $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$

$$\Rightarrow (3x+a) \begin{vmatrix} 1 & 1 & 1 \\ x & a & x \\ x & x & a \end{vmatrix} = 0$$

Expanding along R_1

$$\Rightarrow a^2(3x+a)=0$$

But $a \neq 0$.

Hence, $3x+a=0$

$$\Rightarrow x = -\frac{a}{3}$$

6. Prove that
$$\begin{vmatrix} a^2 & bc & ac+c^2 \\ a^2+ab & b^2 & ac \\ ab & b^2+bc & c^2 \end{vmatrix} = 4a^2b^2c^2.$$

Ans: Given,
$$\Delta = \begin{vmatrix} a^2 & bc & ac+c^2 \\ a^2+ab & b^2 & ac \\ ab & b^2+bc & c^2 \end{vmatrix}$$

Taking out a, b and c from C_1 , C_2 and C_3 .

$$\Delta = abc \begin{vmatrix} a & c & a+c \\ a+b & b & a \\ b & b+c & c \end{vmatrix}$$

Applying the row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$

$$\Delta = abc \begin{vmatrix} a & c & a+c \\ b & b-c & -c \\ b-a & b & -a \end{vmatrix}$$

Applying $R_2 \rightarrow R_2 + R_1$

$$\Delta = abc \begin{vmatrix} a & c & a+c \\ a+b & b & a \\ b-a & b & -a \end{vmatrix}$$

Applying $R_3 \rightarrow R_3 + R_2$

$$\Delta = abc \begin{vmatrix} a & c & a+c \\ a+b & b & a \\ 2b & 2b & 0 \end{vmatrix}$$

$$\Rightarrow \Delta = 2ab^2c \begin{vmatrix} a & c & a+c \\ a+b & b & a \\ 1 & 1 & 0 \end{vmatrix}$$

Applying $C_2 \rightarrow C_2 - C_1$

Expanding along R_3 ,

$$\Delta = 2ab^2c [a(c-a) + a(a+c)]$$

$$\Rightarrow \Delta = 2ab^2c [ac - a^2 + a^2 + ac]$$

$$\Rightarrow \Delta = 2ab^2c (2ac)$$

$$\Rightarrow \Delta = 4a^2b^2c^2$$

Hence proved.

7. If $A^{-1} = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$, find $(AB)^{-1}$.

Ans: The below result will be used for simplification,

$$(AB)^{-1} = B^{-1}A^{-1}$$

Finding inverse of matrix B. so, the determinant is

$$|B| = 1(3-0) - 2(-1-0) - 2(2-0)$$

$$\Rightarrow |B| = 3 + 2 - 4$$

$$\therefore |B| = 1$$

Now finding the cofactors,

$$B_{11} = (-1)^{1+1} \begin{vmatrix} 3 & 0 \\ -2 & 1 \end{vmatrix} \Rightarrow B_{11} = 3$$

$$B_{12} = (-1)^{1+2} \begin{vmatrix} -1 & 0 \\ 0 & 1 \end{vmatrix} \Rightarrow B_{12} = 1$$

$$B_{13} = (-1)^{1+3} \begin{vmatrix} -1 & 3 \\ 0 & -2 \end{vmatrix} \Rightarrow B_{13} = 2$$

$$B_{21} = (-1)^{2+1} \begin{vmatrix} 2 & -2 \\ -2 & 1 \end{vmatrix} \Rightarrow B_{21} = -(2 - 4) = 2$$

$$B_{22} = (-1)^{2+2} \begin{vmatrix} 1 & -2 \\ 0 & 1 \end{vmatrix} \Rightarrow B_{22} = 1$$

$$B_{23} = (-1)^{2+3} \begin{vmatrix} 1 & 2 \\ 0 & -2 \end{vmatrix} \Rightarrow B_{23} = 2$$

$$B_{31} = (-1)^{3+1} \begin{vmatrix} 2 & -2 \\ 3 & 0 \end{vmatrix} \Rightarrow B_{31} = 6$$

$$B_{32} = (-1)^{3+2} \begin{vmatrix} 1 & -2 \\ -1 & 0 \end{vmatrix} \Rightarrow B_{32} = 2$$

$$B_{33} = (-1)^{3+3} \begin{vmatrix} 1 & 2 \\ -1 & 3 \end{vmatrix} \Rightarrow B_{33} = (3 + 2) = 5$$

The cofactor matrix is $\begin{bmatrix} 3 & 1 & 2 \\ 2 & 1 & 2 \\ 6 & 2 & 5 \end{bmatrix}$. The adjoint will be the transpose of cofactor matrix.

$$\text{adj}(B) = \begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix}$$

The inverse is given by $B^{-1} = \frac{\text{adj}(B)}{|B|}$. So,

$$B^{-1} = \begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix}$$

Now, it is already given that $A^{-1} = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$.

So, we can compute $(AB)^{-1} = B^{-1}A^{-1}$ as below,

$$\begin{aligned} (AB)^{-1} &= \begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix} \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix} \\ \Rightarrow (AB)^{-1} &= \begin{bmatrix} 9-30+30 & -3+12-12 & 3-10+12 \\ 3-15+10 & -1+6-4 & 1-5+4 \\ 6-30+25 & -2+12-10 & 2-10+10 \end{bmatrix} \end{aligned}$$

$$\therefore (AB)^{-1} = \begin{bmatrix} 9 & -3 & 5 \\ -2 & 1 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

8. Let $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 5 \end{bmatrix}$ verify that

i. $[\text{adj}A]^{-1} = \text{adj}(A^{-1})$

Ans: Given, $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 5 \end{bmatrix}$

$$\Rightarrow |A| = 1(15-1) - 2(10-1) + 1(2-3)$$

$$\Rightarrow |A| = 14 - 18 - 1$$

$$\therefore |A| = -5$$

Thus,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = 14$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = -9$$

$$\Rightarrow A_{13} = (-1)^{1+3} M_{13} = (-1)^4 M_{13}$$

$$\Rightarrow A_{13} = -1$$

Similarly,

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = -9$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = 4$$

$$\Rightarrow A_{23} = (-1)^{2+3} M_{23} = (-1)^5 M_{23}$$

$$\Rightarrow A_{23} = 1$$

And

$$\Rightarrow A_{31} = (-1)^{3+1} M_{31} = (-1)^4 M_{31}$$

$$\Rightarrow A_{31} = -1$$

$$\Rightarrow A_{32} = (-1)^{3+2} M_{32} = (-1)^5 M_{32}$$

$$\Rightarrow A_{32} = 1$$

$$\Rightarrow A_{33} = (-1)^{3+3} M_{33} = (-1)^6 M_{33}$$

$$\Rightarrow A_{33} = -1.$$

Cofactor matrix is $\begin{bmatrix} 14 & -9 & -1 \\ -9 & 4 & 1 \\ -1 & 1 & -1 \end{bmatrix}.$

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\Rightarrow \text{adj}A = \begin{bmatrix} 14 & -9 & -1 \\ -9 & 4 & 1 \\ -1 & 1 & -1 \end{bmatrix}^T$$

$$\therefore \text{adj}A = \begin{bmatrix} 14 & -9 & -1 \\ -9 & 4 & 1 \\ -1 & 1 & -1 \end{bmatrix}$$

Let us denote the adjoint of A as B. So, $B = \begin{bmatrix} 14 & -9 & -1 \\ -9 & 4 & 1 \\ -1 & 1 & -1 \end{bmatrix}$.

The inverse of A is given by

$$A^{-1} = \frac{1}{|A|}(\text{adj}A)$$

$$\Rightarrow A^{-1} = -\frac{1}{5} \begin{bmatrix} 14 & -9 & -1 \\ -9 & 4 & 1 \\ -1 & 1 & -1 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{5} \begin{bmatrix} -14 & 9 & 1 \\ 9 & -4 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

Now, we have to verify $[\text{adj}A]^{-1} = \text{adj}(A^{-1})$.

Let us compute the RHS first, i.e. the adjoint of $A^{-1} = \frac{1}{5} \begin{bmatrix} -14 & 9 & 1 \\ 9 & -4 & -1 \\ 1 & -1 & 1 \end{bmatrix}$ or

$$A^{-1} = \begin{bmatrix} -\frac{14}{5} & \frac{9}{5} & \frac{1}{5} \\ \frac{9}{5} & -\frac{4}{5} & -\frac{1}{5} \\ \frac{1}{5} & -\frac{1}{5} & \frac{1}{5} \end{bmatrix}.$$

Thus,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = -\frac{1}{5}$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = -\frac{2}{5}$$

$$\Rightarrow A_{13} = (-1)^{1+3} M_{13} = (-1)^4 M_{13}$$

$$\Rightarrow A_{13} = -\frac{1}{5}$$

Similarly,

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = -\frac{2}{5}$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = -\frac{3}{5}$$

$$\Rightarrow A_{23} = (-1)^{2+3} M_{23} = (-1)^5 M_{23}$$

$$\Rightarrow A_{23} = -\frac{1}{5}$$

And

$$\Rightarrow A_{31} = (-1)^{3+1} M_{31} = (-1)^4 M_{31}$$

$$\Rightarrow A_{31} = -\frac{1}{5}$$

$$\Rightarrow A_{32} = (-1)^{3+2} M_{32} = (-1)^5 M_{32}$$

$$\Rightarrow A_{32} = -\frac{1}{5}$$

$$\Rightarrow A_{33} = (-1)^{3+3} M_{33} = (-1)^6 M_{33}$$

$$\Rightarrow A_{33} = -1.$$

Cofactor matrix is $\begin{bmatrix} -\frac{1}{5} & -\frac{2}{5} & -\frac{1}{5} \\ -\frac{2}{5} & -\frac{3}{5} & -\frac{1}{5} \\ -\frac{1}{5} & -\frac{1}{5} & -1 \end{bmatrix}$. We know that adjoint of a matrix is the

transpose of its cofactor matrix.

$$\Rightarrow \text{adj} A^{-1} = \begin{bmatrix} -\frac{1}{5} & -\frac{2}{5} & -\frac{1}{5} \\ -\frac{2}{5} & -\frac{3}{5} & -\frac{1}{5} \\ -\frac{1}{5} & -\frac{1}{5} & -1 \end{bmatrix}^T$$

$$\therefore \text{adj} A^{-1} = \begin{bmatrix} -\frac{1}{5} & -\frac{2}{5} & -\frac{1}{5} \\ -\frac{2}{5} & -\frac{3}{5} & -\frac{1}{5} \\ -\frac{1}{5} & -\frac{1}{5} & -1 \end{bmatrix}$$

Next, moving to the LHS, i.e. the inverse of adjoint of A. We have adjoint of A as matrix B,

$$B = \begin{bmatrix} 14 & -9 & -1 \\ -9 & 4 & 1 \\ -1 & 1 & -1 \end{bmatrix}$$

Determinant of B is

$$|B| = [14(-4-1) + 9(9+1) - 1(-9+4)]$$

$$\Rightarrow |B| = [-70 + 90 + 5]$$

$$\therefore |B| = 25$$

Now the cofactors,

Thus,

$$\Rightarrow B_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow B_{11} = -5$$

$$\Rightarrow B_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow B_{12} = -10$$

$$\Rightarrow B_{13} = (-1)^{1+3} M_{13} = (-1)^4 M_{13}$$

$$\Rightarrow B_{13} = -5$$

Similarly,

$$\Rightarrow B_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow B_{21} = -10$$

$$\Rightarrow B_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow B_{22} = -15$$

$$\Rightarrow B_{23} = (-1)^{2+3} M_{23} = (-1)^5 M_{23}$$

$$\Rightarrow B_{23} = -5$$

And

$$\Rightarrow B_{31} = (-1)^{3+1} M_{31} = (-1)^4 M_{31}$$

$$\Rightarrow B_{31} = -5$$

$$\Rightarrow B_{32} = (-1)^{3+2} M_{32} = (-1)^5 M_{32}$$

$$\Rightarrow B_{32} = -5$$

$$\Rightarrow B_{33} = (-1)^{3+3} M_{33} = (-1)^6 M_{33}$$

$$\Rightarrow B_{33} = -25.$$

Cofactor matrix is $\begin{bmatrix} -5 & -10 & -5 \\ -10 & -15 & -5 \\ -5 & -5 & -25 \end{bmatrix}.$

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\Rightarrow \text{adj}B = \begin{bmatrix} -5 & -10 & -5 \\ -10 & -15 & -5 \\ -5 & -5 & -25 \end{bmatrix}^T$$

$$\therefore \text{adj}B = \begin{bmatrix} -5 & -10 & -5 \\ -10 & -15 & -5 \\ -5 & -5 & -25 \end{bmatrix}$$

Now the inverse is found as

$$B^{-1} = \frac{1}{|B|}(\text{adj}B)$$

$$\Rightarrow B^{-1} = \frac{1}{25} \begin{bmatrix} -5 & -10 & -5 \\ -10 & -15 & -5 \\ -5 & -5 & -25 \end{bmatrix}$$

$$\therefore B^{-1} = \begin{bmatrix} -\frac{1}{5} & -\frac{2}{5} & -\frac{1}{5} \\ -\frac{2}{5} & -\frac{3}{5} & -\frac{1}{5} \\ -\frac{1}{5} & -\frac{1}{5} & -1 \end{bmatrix}$$

$$\text{So, LHS is } [\text{adj}A]^{-1} = \begin{bmatrix} -\frac{1}{5} & -\frac{2}{5} & -\frac{1}{5} \\ -\frac{2}{5} & -\frac{3}{5} & -\frac{1}{5} \\ -\frac{1}{5} & -\frac{1}{5} & -1 \end{bmatrix}.$$

Since LHS=RHS, it is verified that $[\text{adj}A]^{-1} = \text{adj}(A^{-1})$.

ii. $(A^{-1})^{-1} = A$

Ans:

$$\text{Since } A^{-1} = \begin{bmatrix} -\frac{14}{5} & \frac{9}{5} & \frac{1}{5} \\ \frac{9}{5} & -\frac{4}{5} & -\frac{1}{5} \\ \frac{1}{5} & -\frac{1}{5} & \frac{1}{5} \end{bmatrix} \text{ and } \text{adj}A^{-1} = \begin{bmatrix} -\frac{1}{5} & -\frac{2}{5} & -\frac{1}{5} \\ -\frac{2}{5} & -\frac{3}{5} & -\frac{1}{5} \\ -\frac{1}{5} & -\frac{1}{5} & -1 \end{bmatrix}, \text{ thus,}$$

$$\Rightarrow |A^{-1}| = \begin{bmatrix} -\frac{14}{5} \left(-\frac{4}{25} - \frac{1}{25} \right) - \frac{9}{5} \left(\frac{9}{25} + \frac{1}{25} \right) + \frac{1}{5} \left(-\frac{9}{25} + \frac{4}{25} \right) \end{bmatrix}$$

$$\Rightarrow |A^{-1}| = \begin{bmatrix} \frac{70}{125} - \frac{90}{125} - \frac{5}{125} \end{bmatrix}$$

$$\therefore |A^{-1}| = -\frac{1}{5}$$

Also, we know that an inverse of a matrix is given by:

$$\Rightarrow (A^{-1})^{-1} = \frac{\text{adj}A^{-1}}{|A^{-1}|}$$

$$\Rightarrow (A^{-1})^{-1} = -5 \begin{bmatrix} -\frac{1}{5} & -\frac{2}{5} & -\frac{1}{5} \\ -\frac{2}{5} & -\frac{3}{5} & -\frac{1}{5} \\ -\frac{1}{5} & -\frac{1}{5} & -1 \end{bmatrix}$$

$$\Rightarrow (A^{-1})^{-1} = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\Rightarrow (A^{-1})^{-1} = A$$

Hence verified that $(A^{-1})^{-1} = A$.

9. Evaluate
$$\begin{vmatrix} x & y & x+y \\ y & x+y & x \\ x+y & x & y \end{vmatrix}$$

Ans: Given, $\Delta = \begin{vmatrix} x & y & x+y \\ y & x+y & x \\ x+y & x & y \end{vmatrix}$

Applying $R_1 \rightarrow R_1 + R_2 + R_3$

$$\Delta = \begin{vmatrix} 2(x+y) & 2(x+y) & 2(x+y) \\ y & x+y & x \\ x+y & x & y \end{vmatrix}$$

Taking $2(x+y)$ common from R_1

$$\Rightarrow \Delta = 2(x+y) \begin{vmatrix} 1 & 1 & 1 \\ y & x+y & x \\ x+y & x & y \end{vmatrix}$$

Applying the row operations $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$

$$\Delta = 2(x+y) \begin{vmatrix} 1 & 0 & 0 \\ y & x & x-y \\ x+y & -y & -x \end{vmatrix}$$

Expanding along R_1

$$\Rightarrow \Delta = 2(x+y) [1((x \times (-x)) - (-y(x-y))) + 0 + 0]$$

$$\Rightarrow \Delta = 2(x+y) \left[1 \left((-x^2) - (-yx + y^2) \right) \right]$$

$$\Rightarrow \Delta = 2(x+y) \left[-x^2 + xy - y^2 \right]$$

$$\Rightarrow \Delta = -2(x+y) \left[x^2 - xy + y^2 \right]$$

Applying the identity,

$$\therefore \Delta = -2 \left[x^3 + y^3 \right]$$

10. Evaluate
$$\begin{vmatrix} 1 & x & y \\ 1 & x+y & y \\ 1 & x & x+y \end{vmatrix}$$

Ans: Given,
$$\Delta = \begin{vmatrix} 1 & x & y \\ 1 & x+y & y \\ 1 & x & x+y \end{vmatrix}$$

Applying the row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$

$$\Rightarrow \Delta = \begin{vmatrix} 1 & x & y \\ 0 & y & 0 \\ 0 & 0 & x \end{vmatrix}$$

Expanding along C_1

$$\Rightarrow \Delta = 1(xy - 0)$$

$$\therefore \Delta = xy$$

11. Using properties of determinants, prove that:

$$\begin{vmatrix} \alpha & \alpha^2 & \beta+\gamma \\ \beta & \beta^2 & \gamma+\alpha \\ \gamma & \gamma^2 & \alpha+\beta \end{vmatrix} = (\beta-\gamma)(\gamma-\alpha)(\alpha-\beta)(\alpha+\beta+\gamma)$$

Ans: Given, $\Delta = \begin{vmatrix} \alpha & \alpha^2 & \beta+\gamma \\ \beta & \beta^2 & \gamma+\alpha \\ \gamma & \gamma^2 & \alpha+\beta \end{vmatrix}$

Applying the row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$

$$\Rightarrow \Delta = \begin{vmatrix} \alpha & \alpha^2 & \beta+\gamma \\ \beta-\alpha & \beta^2-\alpha^2 & \alpha-\beta \\ \gamma-\alpha & \gamma^2-\alpha^2 & \alpha-\gamma \end{vmatrix}$$

$$\Rightarrow \Delta = (\beta-\alpha)(\gamma-\alpha) \begin{vmatrix} \alpha & \alpha^2 & \beta+\gamma \\ 1 & \beta+\alpha & -1 \\ 1 & \gamma+\alpha & -1 \end{vmatrix}$$

Applying $R_3 \rightarrow R_3 - R_2$

$$\Rightarrow \Delta = (\beta-\alpha)(\gamma-\alpha) \begin{vmatrix} \alpha & \alpha^2 & \beta+\gamma \\ 1 & \beta+\alpha & -1 \\ 0 & \gamma-\beta & 0 \end{vmatrix}$$

Expanding along R_3

$$\Rightarrow \Delta = (\beta-\alpha)(\gamma-\alpha) [-(\gamma-\beta)(-\alpha-\beta-\gamma)]$$

$$\Rightarrow \Delta = (\beta-\alpha)(\gamma-\alpha)(\gamma-\beta)(\alpha+\beta+\gamma)$$

$$\Rightarrow \Delta = (\beta-\gamma)(\gamma-\alpha)(\alpha-\beta)(\alpha+\beta+\gamma)$$

Hence proved.

12. Using properties of determinants, prove that:

$$\begin{vmatrix} x & x^2 & 1+px^3 \\ y & y^2 & 1+py^3 \\ z & z^2 & 1+pz^3 \end{vmatrix} = (1+pxyz)(x-y)(y-z)(z-x)$$

Ans: Given, $\Delta = \begin{vmatrix} x & x^2 & 1+px^3 \\ y & y^2 & 1+py^3 \\ z & z^2 & 1+pz^3 \end{vmatrix}$

Applying the row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$

$$\Rightarrow \Delta = \begin{vmatrix} x & x^2 & 1+px^3 \\ y-x & y^2-x & p(y^3-x^3) \\ z-x & z^2-x & p(z^3-x^3) \end{vmatrix}$$

$$\Rightarrow \Delta = (y-x)(z-x) \begin{vmatrix} x & x^2 & 1+px^3 \\ 1 & y-x & p(y^2-x^2+xy) \\ 1 & z-x & p(z^2-x^2+xz) \end{vmatrix}$$

Applying $R_3 \rightarrow R_3 - R_2$

$$\Rightarrow \Delta = (y-x)(z-x) \begin{vmatrix} x & x^2 & 1+px^3 \\ 1 & y-x & p(y^2-x^2+xy) \\ 0 & z-y & p(z-y)(x+y+z) \end{vmatrix}$$

Taking $(z-y)$ from R_3

$$\Rightarrow \Delta = (y-x)(z-x)(z-y) \begin{vmatrix} x & x^2 & 1+px^3 \\ 1 & y+x & p(y^2-x^2+xy) \\ 0 & 1 & p(x+y+z) \end{vmatrix}$$

Expanding along R_3

$$\Rightarrow \Delta = (x-y)(y-z)(z-x) \left[-1 \left(xp(y^2-x^2+xy) - (1+px^3) \right) + p(x+y+z)(x(y+x) - x^2) \right]$$

$$\Rightarrow \Delta = (x-y)(y-z)(z-x) \left[-1 \left(xpy^2 - px^3 + px^2y - 1 - px^3 \right) + p(x+y+z)(xy + x^2 - x^2) \right]$$

$$\Rightarrow \Delta = (x-y)(y-z)(z-x) \left[-xpy^2 + px^3 - px^2y + 1 - px^3 + pxy(x+y+z) \right]$$

$$\Rightarrow \Delta = (x-y)(y-z)(z-x) \left[-xpy^2 + px^3 - px^2y + 1 - px^3 + px^2y + pxy^2 + pxyz \right]$$

$$\Rightarrow \Delta = (x-y)(y-z)(z-x) [1 + pxyz]$$

Hence proved.

13. Using properties of determinants, prove that:

$$\begin{vmatrix} 3a & -a+b & -a+c \\ -b+a & 3b & -b+c \\ -c+a & -c+b & 3c \end{vmatrix} = 3(a+b+c)(ab+ba+ca)$$

Ans: Given, $\Delta = \begin{vmatrix} 3a & -a+b & -a+c \\ -b+a & 3b & -b+c \\ -c+a & -c+b & 3c \end{vmatrix}$

Applying the column operation, $C_1 \rightarrow C_1 + C_2 + C_3$

$$\Rightarrow \Delta = \begin{vmatrix} a+b+c & -a+b & -a+c \\ a+b+c & 3b & -b+c \\ a+b+c & -c+b & 3c \end{vmatrix}$$

$$\Rightarrow \Delta = (a+b+c) \begin{vmatrix} 1 & -a+b & -a+c \\ 1 & 3b & -b+c \\ 1 & -c+b & 3c \end{vmatrix}$$

Applying $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$

$$\Rightarrow \Delta = (a+b+c) \begin{vmatrix} 1 & -a+b & -a+c \\ 0 & 2b+a & a-b \\ 0 & a-c & 2c+a \end{vmatrix}$$

Expanding along C_1

$$\Rightarrow \Delta = (a+b+c) [(2b+a)(2c+a) - (a-b)(a-c)]$$

$$\Rightarrow \Delta = (a+b+c) [4bc + 2ab + 2ac + a^2 - a^2 + ac + ba - bc]$$

$$\Rightarrow \Delta = (a+b+c)(3ab + 3bc + 3ac)$$

$$\Rightarrow \Delta = 3(a+b+c)(ab + bc + ca)$$

$$\therefore \begin{vmatrix} 3a & -a+b & -a+c \\ -b+a & 3b & -b+c \\ -c+a & -c+b & 3c \end{vmatrix} = 3(a+b+c)(ab + ba + ca)$$

Hence proved.

14. Using properties of determinants, prove this:

$$\begin{vmatrix} 1 & 1+p & 1+p+q \\ 2 & 3+2p & 4+3p+2q \\ 3 & 6+3p & 10+6p+3q \end{vmatrix} = 1$$

Ans: Given, $\Delta = \begin{vmatrix} 1 & 1+p & 1+p+q \\ 2 & 3+2p & 4+3p+2q \\ 3 & 6+3p & 10+6p+3q \end{vmatrix}$

Applying $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 3R_1$

$$\Rightarrow \Delta = \begin{vmatrix} 1 & 1+p & 1+p+q \\ 0 & 1 & 2+p \\ 0 & 3 & 7+3p \end{vmatrix}$$

Applying $R_3 \rightarrow R_3 - 3R_2$, we have:

$$\Rightarrow \Delta = \begin{vmatrix} 1 & 1+p & 1+p+q \\ 0 & 1 & 2+p \\ 0 & 0 & 1 \end{vmatrix}$$

Expanding along C_1

$$\Rightarrow \Delta = 1 \begin{vmatrix} 1 & 2+p \\ 0 & 1 \end{vmatrix}$$

$$\Rightarrow \Delta = 1(1-0)$$

$$\Rightarrow \Delta = 1$$

$$\therefore \begin{vmatrix} 1 & 1+p & 1+p+q \\ 2 & 3+2p & 4+3p+2q \\ 3 & 6+3p & 10+6p+3q \end{vmatrix} = 1$$

15. Using properties of determinants, prove that:

$$\begin{vmatrix} \sin \alpha & \cos \alpha & \cos(\alpha + \delta) \\ \sin \beta & \cos \beta & \cos(\beta + \delta) \\ \sin \gamma & \cos \gamma & \cos(\gamma + \delta) \end{vmatrix} = 0$$

Ans: Given, $\Delta = \begin{vmatrix} \sin \alpha & \cos \alpha & \cos(\alpha + \delta) \\ \sin \beta & \cos \beta & \cos(\beta + \delta) \\ \sin \gamma & \cos \gamma & \cos(\gamma + \delta) \end{vmatrix}$

$$\Rightarrow \Delta = \frac{1}{\sin \delta \cos \delta} \begin{vmatrix} \sin \alpha \sin \delta & \cos \alpha \cos \delta & \cos \alpha \cos \delta - \sin \alpha \sin \delta \\ \sin \beta \sin \delta & \cos \beta \cos \delta & \cos \beta \cos \delta - \sin \beta \sin \delta \\ \sin \gamma \sin \delta & \cos \gamma \cos \delta & \cos \gamma \cos \delta - \sin \gamma \sin \delta \end{vmatrix}$$

Applying $C_1 \rightarrow C_1 + C_3$

$$\Rightarrow \Delta = \frac{1}{\sin \delta \cos \delta} \begin{vmatrix} \cos \alpha \cos \delta & \cos \alpha \cos \delta & \cos \alpha \cos \delta - \sin \alpha \sin \delta \\ \cos \beta \cos \delta & \cos \beta \cos \delta & \cos \beta \cos \delta - \sin \beta \sin \delta \\ \cos \gamma \cos \delta & \cos \gamma \cos \delta & \cos \gamma \cos \delta - \sin \gamma \sin \delta \end{vmatrix}$$

Since, two columns C_1 and C_2 are identical.

$$\Rightarrow \Delta = 0$$

$$\therefore \begin{vmatrix} \sin \alpha & \cos \alpha & \cos(\alpha + \delta) \\ \sin \beta & \cos \beta & \cos(\beta + \delta) \\ \sin \gamma & \cos \gamma & \cos(\gamma + \delta) \end{vmatrix} = 0$$

Hence proved.

16. Solve the system of the following equations

$$\frac{2}{x} + \frac{3}{y} + \frac{10}{z} = 4$$

$$\frac{4}{x} + \frac{6}{y} + \frac{5}{z} = 1$$

$$\frac{6}{x} + \frac{9}{y} + \frac{20}{z} = 2$$

Ans: Given equations,

$$\frac{2}{x} + \frac{3}{y} + \frac{10}{z} = 4$$

$$\frac{4}{x} + \frac{6}{y} + \frac{5}{z} = 1$$

$$\frac{6}{x} + \frac{9}{y} + \frac{20}{z} = 2$$

Let $\frac{1}{x} = p$, $\frac{1}{y} = q$ and $\frac{1}{z} = r$

Then the given system of equations becomes:

$$2p + 3q + 10r = 4$$

$$4p - 6q + 5r = 1$$

$$6p + 9q + 20r = 2$$

Let $A = \begin{bmatrix} 2 & 3 & 10 \\ 4 & -6 & 5 \\ 6 & 9 & -20 \end{bmatrix}$, $X = \begin{bmatrix} p \\ q \\ r \end{bmatrix}$ and $B = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$ such that, this system can be written in

the form of $AX = B$.

Now,

$$\Rightarrow |A| = 2(120 - 45) - 3(-80 - 30) + 10(36 + 36)$$

$$\Rightarrow |A| = 150 + 330 + 720$$

$$\Rightarrow |A| = 1200$$

Thus, A is a non-singular matrix.

$\therefore$ Its inverse exists.

So,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 \begin{vmatrix} -6 & 5 \\ 9 & -20 \end{vmatrix}$$

$$\Rightarrow A_{11} = 75$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 \begin{vmatrix} 4 & 5 \\ 6 & -20 \end{vmatrix}$$

$$\Rightarrow A_{12} = 110$$

$$\Rightarrow A_{13} = (-1)^{1+3} M_{13} = (-1)^4 \begin{vmatrix} 4 & -6 \\ 6 & 9 \end{vmatrix}$$

$$\Rightarrow A_{13} = 72$$

Similarly,

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 \begin{vmatrix} 3 & 10 \\ 9 & -20 \end{vmatrix}$$

$$\Rightarrow A_{21} = 150$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 \begin{vmatrix} 2 & 10 \\ 6 & -20 \end{vmatrix}$$

$$\Rightarrow A_{22} = 100$$

$$\Rightarrow A_{23} = (-1)^{2+3} M_{23} = (-1)^5 \begin{vmatrix} 2 & 3 \\ 6 & 9 \end{vmatrix}$$

$$\Rightarrow A_{23}=0$$

and

$$\Rightarrow A_{31}=(-1)^{3+1}M_{31}=(-1)^4\begin{vmatrix} 3 & 10 \\ -6 & 5 \end{vmatrix}$$

$$\Rightarrow A_{31}=75$$

$$\Rightarrow A_{32}=(-1)^{3+2}M_{32}=(-1)^5\begin{vmatrix} 2 & 10 \\ 4 & 5 \end{vmatrix}$$

$$\Rightarrow A_{32}=30$$

$$\Rightarrow A_{33}=(-1)^{3+3}M_{33}=(-1)^6\begin{vmatrix} 2 & 3 \\ 4 & -6 \end{vmatrix}$$

$$\Rightarrow A_{33}=-24.$$

The inverse of a matrix is given by:

$$\Rightarrow A^{-1}=\frac{1}{|A|}(\text{adj}A)$$

$$\therefore A^{-1}=\frac{1}{1200}\begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix}$$

Now, $X=A^{-1}B$

$$\Rightarrow \begin{bmatrix} p \\ q \\ r \end{bmatrix}=\frac{1}{1200}\begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix}\begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \frac{1}{1200} \begin{bmatrix} 300+150+150 \\ 440-100+60 \\ 288+0-48 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \frac{1}{1200} \begin{bmatrix} 600 \\ 400 \\ 240 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{3} \\ \frac{1}{5} \end{bmatrix}$$

$$\therefore P = \frac{1}{2}, q = \frac{1}{3} \text{ and } r = \frac{1}{5}$$

Thus $x=2$, $y=3$ and $z=5$.

17. Choose the correct answer.

If a, b, c are in A.P , then the determinant

$$\begin{vmatrix} x+2 & x+3 & x+2a \\ x+3 & x+4 & x+2b \\ x+4 & x+5 & x+2c \end{vmatrix}$$

- A. 0
- B. 1
- C. X
- D. 2X

Ans: Given, $\Delta = \begin{vmatrix} x+2 & x+3 & x+2a \\ x+3 & x+4 & x+2b \\ x+4 & x+5 & x+2c \end{vmatrix}$

Since a, b, and c are in A.P. $2b=a+c$,

$$\Rightarrow \Delta = \begin{vmatrix} x+2 & x+3 & x+2a \\ x+3 & x+4 & x+(a+c) \\ x+4 & x+5 & x+2c \end{vmatrix}$$

Applying $R_1 \rightarrow R_1 - R_2$ and $R_3 \rightarrow R_3 - R_2$

$$\Rightarrow \Delta = \begin{vmatrix} -1 & -1 & a-c \\ x+3 & x+4 & x+(a+c) \\ 1 & 1 & c-a \end{vmatrix}$$

Applying $R_1 \rightarrow R_1 - R_3$

$$\Rightarrow \Delta = \begin{vmatrix} 0 & 0 & 0 \\ x+3 & x+4 & x+a+c \\ 1 & 1 & c-a \end{vmatrix}$$

$$\therefore \Delta = 0$$

18. Choose the correct answer.

If X, Y, Z are nonzero real numbers, then the inverse of matrix $A = \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}$

A. $\begin{bmatrix} x^{-1} & 0 & 0 \\ 0 & y^{-1} & 0 \\ 0 & 0 & z^{-1} \end{bmatrix}$

B. $xyz \begin{bmatrix} x^{-1} & 0 & 0 \\ 0 & y^{-1} & 0 \\ 0 & 0 & z^{-1} \end{bmatrix}$

C. $\frac{1}{xyz} \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}$

D. $\frac{1}{xyz} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Ans: Given, $A = \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}$

$$\Rightarrow |A| = x(yz - 0)$$

$$\therefore |A| = xyz \neq 0$$

Thus,

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = (-1)^2 M_{11}$$

$$\Rightarrow A_{11} = yz$$

$$\Rightarrow A_{12} = (-1)^{1+2} M_{12} = (-1)^3 M_{12}$$

$$\Rightarrow A_{12} = 0$$

$$\Rightarrow A_{13} = (-1)^{1+3} M_{13} = (-1)^4 M_{13}$$

$$\Rightarrow A_{13} = 0$$

Similarly,

$$\Rightarrow A_{21} = (-1)^{2+1} M_{21} = (-1)^3 M_{21}$$

$$\Rightarrow A_{21} = 0$$

$$\Rightarrow A_{22} = (-1)^{2+2} M_{22} = (-1)^4 M_{22}$$

$$\Rightarrow A_{22} = xz$$

$$\Rightarrow A_{23} = (-1)^{2+3} M_{23} = (-1)^5 M_{23}$$

$$\Rightarrow A_{23} = 0$$

and

$$\Rightarrow A_{31} = (-1)^{3+1} M_{31} = (-1)^4 M_{31}$$

$$\Rightarrow A_{31} = 0$$

$$\Rightarrow A_{32} = (-1)^{3+2} M_{32} = (-1)^5 M_{32}$$

$$\Rightarrow A_{32} = 0$$

$$\Rightarrow A_{33} = (-1)^{3+3} M_{33} = (-1)^6 M_{33}$$

$$\Rightarrow A_{33} = xy.$$

Cofactor matrix is $\begin{bmatrix} yz & 0 & 0 \\ 0 & xz & 0 \\ 0 & 0 & xy \end{bmatrix}.$

We know that adjoint of a matrix is the transpose of its cofactor matrix.

$$\Rightarrow \text{adj}A = \begin{bmatrix} yz & 0 & 0 \\ 0 & xz & 0 \\ 0 & 0 & xy \end{bmatrix}^T$$

$$\therefore \text{adj}A = \begin{bmatrix} yz & 0 & 0 \\ 0 & xz & 0 \\ 0 & 0 & xy \end{bmatrix}$$

The inverse of a matrix is given by:

$$\Rightarrow A^{-1} = \frac{1}{|A|}(\text{adj}A)$$

$$\Rightarrow A^{-1} = \frac{1}{xyz} \begin{bmatrix} yz & 0 & 0 \\ 0 & xz & 0 \\ 0 & 0 & xy \end{bmatrix}$$

$$\Rightarrow A^{-1} = \begin{bmatrix} \frac{yz}{xyz} & 0 & 0 \\ 0 & \frac{xz}{xyz} & 0 \\ 0 & 0 & \frac{xy}{xyz} \end{bmatrix}$$

$$\Rightarrow A^{-1} = \begin{bmatrix} \frac{1}{x} & 0 & 0 \\ 0 & \frac{1}{y} & 0 \\ 0 & 0 & \frac{1}{z} \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} x^{-1} & 0 & 0 \\ 0 & y^{-1} & 0 \\ 0 & 0 & z^{-1} \end{bmatrix}$$

Thus, **A.** $\begin{bmatrix} x^{-1} & 0 & 0 \\ 0 & y^{-1} & 0 \\ 0 & 0 & z^{-1} \end{bmatrix}$ is the correct answer.

19. Choose the correct answer.

Let **A** = $\begin{bmatrix} 1 & \sin \theta & 1 \\ -\sin \theta & 1 & \sin \theta \\ -1 & -\sin \theta & 1 \end{bmatrix}$, where $0 \leq \theta \leq 2\pi$, then

A. $\text{Det}(\mathbf{A})=0$

B. $\text{Det}(\mathbf{A}) \in (2, \infty)$

C. $\text{Det}(\mathbf{A}) \in (2, 4)$

D. $\text{Det}(\mathbf{A}) \in [2, 4]$

Ans: Given, **A** = $\begin{bmatrix} 1 & \sin \theta & 1 \\ -\sin \theta & 1 & \sin \theta \\ -1 & -\sin \theta & 1 \end{bmatrix}$

$$\therefore |A| = 1(1 + \sin^2 \theta) - \sin \theta(-\sin \theta + \sin \theta) + 1(\sin^2 \theta + 1)$$

$$\Rightarrow |A| = 1 + \sin^2 \theta + \sin^2 \theta + 1$$

$$\Rightarrow |A| = 2 + 2\sin^2 \theta$$

$$\Rightarrow |A| = 2(1 + \sin^2 \theta)$$

We know, $0 \leq \sin \theta \leq 1$

$$\Rightarrow 1 \leq 1 + \sin^2 \theta \leq 2$$

$$\Rightarrow 2 \leq 2(1 + \sin^2 \theta) \leq 4$$

Thus, **D.** $\text{Det}(\mathbf{A}) \in [2, 4]$ is the correct answer.